

Econometrics Note (II) *

Yuan-Ting Lee

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1 Basic Regression Analysis with Time Series Data

1.1 The Nature of Time Series Data

An obvious characteristic of time series data that distinguishes them from cross-sectional data is **temporal ordering**. Also, another typical feature is **serial correlation non-independence**. We must recognize that the past can affect the future, but not vice versa. Economic time series satisfy the intuitive requirements for being outcomes of random variables.

Remark. Time series analysis focus on the modeling the *dependence* of a variable on *its own path* and on the *present* and *past values* of other variables.

- A sequence of random variables indexed by time is called a **stochastic process** or a **time series process**.
- A time series data set, we obtain one possible outcome, or realization, of the stochastic process.
- The sample size for a time series data set is the number of time periods over which we observe the variables of interest.

1.2 Examples of Time Series Regression Models

1.2.1 Static Models

A **static model** relating y to z is

$$y_t = \beta_0 + \beta_1 z_t + u_t, \quad t = 1, 2, \dots, n. \quad (1)$$

Since we are modeling a contemporaneous relationship between y and z , “static model” comes from this fact. Usually, a static model is postulated when a change in z at time t is believed to have an immediate effect on y : $\Delta y_t = \beta_1 \Delta z_t$, when $\Delta u_t = 0$. Static regression models are also used when we are interested in knowing the tradeoff between y and z .

Example 1. static Phillips curve

$$inf_t = \beta_0 + \beta_1 unem_t + u_t, \quad (2)$$

where inf_t is the annual inflation rate and $unem_t$ is the annual unemployment rate. This can be used to study the contemporaneous tradeoff between inflation and unemployment.

1.2.2 Finite Distributed Lag Models

In a **finite distributed lag (FDL) model**, we allow one or more variables to affect y with a lag. We consider the model

$$gfr_t = \alpha_0 + \delta_0 pe_t + \delta_1 pe_{t-1} + \delta_2 pe_{t-2} + u_t, \quad (3)$$

where gfr_t is the general fertility rate (children born per 1,000 women of childbearing age) and pe_t is the real dollar value of the personal tax exemption. Equation (3) is an example of the model

$$y_t = \alpha_0 + \delta_0 z_t + \delta_1 z_{t-1} + \delta_2 z_{t-2} + u_t, \quad (4)$$

*Mainly adapted from Jeffrey M. Wooldridge—Introductory Econometrics A Modern Approach, 7e

which is an FDL of *order two*. Suppose that z is a constant, equal to c , in all time periods before time t . At time t , z increases by one unit to $c + 1$ and then reverts to its previous level at time $t + 1$. (That is, the increase in z is temporary.) We can show as follow

$$\dots, z_{t-2} = c, z_{t-1} = c, z_t = c + 1, z_{t+1} = c, z_{t+2} = c, \dots$$

We set the error term in each time period to zero. Then,

$$\begin{aligned} y_{t-1} &= \alpha_0 + \delta_0 c + \delta_1 c + \delta_2 c, \\ y_t &= \alpha_0 + \delta_0(c + 1) + \delta_1 c + \delta_2 c, \\ y_{t+1} &= \alpha_0 + \delta_0 c + \delta_1(c + 1) + \delta_2 c, \\ y_{t+2} &= \alpha_0 + \delta_0 c + \delta_1 c + \delta_2(c + 1), \\ y_{t+3} &= \alpha_0 + \delta_0 c + \delta_1 c + \delta_2 c, \end{aligned}$$

From the first two equations, $y_t - y_{t-1} = \delta_0$, which shows that δ_0 is the immediate change in y due to the one-unit increase in z at time t .

Similarly, $\delta_1 = y_{t+1} - y_t$ is the change in y one period after the temporary change and $\delta_2 = y_{t+2} - y_{t+1}$ is the change in y two periods after the change. At time $t + 3$, y has reverted back to its initial level: $y_{t+3} = y_{t-1}$. This is because we have assumed that only two lags of z appear in (4). When we graph the δ_j as a function of j , we obtain the **lag distribution**, which summarizes the dynamic effect that a temporary increase in z has on y .

If the change in y due to a **permanent** increase in z . Before time t , z equals the constant c . At time t , z increases permanently to $c + 1$: $z_s = c$, $s < t$ and $z_s = c + 1$, $s \geq t$. Setting the errors to zero, we have

$$\begin{aligned} y_{t-1} &= \alpha_0 + \delta_0 c + \delta_1 c + \delta_2 c, \\ y_t &= \alpha_0 + \delta_0(c + 1) + \delta_1 c + \delta_2 c, \\ y_{t+1} &= \alpha_0 + \delta_0(c + 1) + \delta_1(c + 1) + \delta_2 c, \\ y_{t+2} &= \alpha_0 + \delta_0(c + 1) + \delta_1(c + 1) + \delta_2(c + 1). \end{aligned}$$

With the permanent increase in z , after one period, y has increased by $\delta_0 + \delta_1$, and after two periods, y has increased by $\delta_0 + \delta_1 + \delta_2$ which is the **long-run change** in y given a permanent increase in z .

Remark. Transitory shock : effect of a post one-time shock of the current value of Y .

$$\frac{\Delta Y_t}{\Delta Z_{t-s}} = \delta_s$$

δ_s is called the **impact propensity** or **impact multiplier**.

Permanent shock : the explanatory variable permanently increases by one unit

$$\frac{\Delta Y_t}{\Delta Z_{t-q}} + \dots + \frac{\Delta Y}{\Delta Z_t} = \delta_q + \delta_{q-1} + \dots + \delta_0$$

$\delta_q + \delta_{q-1} + \dots + \delta_0$ is called the **long-run propensity (LRP)** or **long-run multiplier**.

An FDL of order q is written as

$$y_t = \alpha_0 + \delta_0 z_t + \delta_1 z_{t-1} + \dots + \delta_q z_{t-q} + u_t. \quad (5)$$

This contains the static model as a special case by setting $\delta_1, \delta_2, \dots, \delta_q$ equal to zero. For estimating a distributed lag model is to test whether z has a lagged effect on y . The impact propensity is always the coefficient on the contemporaneous z , δ_0 . In the general case, the lag distribution can be plotted by graphing the (estimated) δ_j as a function of j .

The LRP is the cumulative effect after all changes have taken place; it is simply the sum of all of the coefficients on the z_{t-j} :

$$\text{LRP} = \delta_0 + \delta_1 + \dots + \delta_q. \quad (6)$$

Interestingly, even when the δ_j cannot be precisely estimated, we can often get good estimates of the LRP.

1.3 Finite Sample Properties of OLS under Classical Assumptions

We focus on the finite sample, or small sample, properties of OLS under standard assumptions.

1.3.1 Unbiasedness of OLS

Assumption TS.1 – Linear in Parameters

The stochastic process $\{(x_{t1}, x_{t2}, \dots, x_{tk}, y_t) : t = 1, 2, \dots, n\}$ follows the linear model

$$y_t = \beta_0 + \beta_1 x_{t1} + \dots + \beta_k x_{tk} + u_t, \quad (7)$$

where $\{u_t : t = 1, 2, \dots, n\}$ is the sequence of errors or disturbances. Here, n is the number of observations (time periods).

In the notation x_{tj} , t denotes the time period, and j is a label to indicate one of the k explanatory variables.

We let $\mathbf{x}_t = (x_{t1}, x_{t2}, \dots, x_{tk})$ denote the set of all independent variables in the equation at time t . Further, $\mathbf{X}$ denotes the collection of all independent variables for all time periods. To think of $\mathbf{X}$ as an array, let it have n rows and k columns. In econometric software, the t^{th} row of $\mathbf{X}$ is $\mathbf{x}_t$, representing all independent variables at time t .

Assumption TS.2 – No Perfect Collinearity

In the sample (and therefore in the underlying time series process), no independent variable is constant nor a perfect linear combination of the others.

Assumption TS.3 – Zero Conditional Mean

For each t , the expected value of the error u_t , given the explanatory variables for *all* time periods, is zero. Mathematically,

$$\mathbb{E}(u_t | \mathbf{X}) = 0, \quad t = 1, 2, \dots, n. \quad (8)$$

Assumption TS.3 implies that the error at time t , u_t , is uncorrelated with each explanatory variable in *every* time period. We require u_t to be uncorrelated with the explanatory variables also dated at time t : in conditional mean terms,

$$\mathbb{E}(u_t | x_{t1}, \dots, x_{tk}) = \mathbb{E}(u_t | x_t) = 0. \quad (9)$$

When (9) holds, we say that the x_{tj} are **contemporaneously exogenous** which implies that u_t and the explanatory variables are contemporaneously uncorrelated: $\text{Corr}(x_{tj}, u_t) = 0$, for all j . If the following holds

$$\mathbb{E}(u_t | \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k) = 0$$

then it's called **sequential exogeneity**. While Assumption TS.3 requires more than contemporaneous exogeneity: u_t must be uncorrelated with x_{sj} , even when $s \neq t$, we say the explanatory variables are **strictly exogenous**.

Assumption TS.3 puts no restriction on correlation in the independent variables or in the u_t across time. It only says that the average value of u_t is unrelated to the independent variables in all time periods.

But, two leading candidates for failure are:

- omitted variables, and
- measurement error in some of the regressors.

In the simple static regression model:

$$y_t = \beta_0 + \beta_1 z_t + u_t,$$

Assumption TS.3 requires not only that u_t and z_t are uncorrelated, but that u_t is also uncorrelated with *past and future* values of z . This has two implications:

1. z can have no lagged effect on y , otherwise we should estimate a distributed lag model.
2. strict exogeneity excludes the possibility that changes in the error term today can cause future changes in z . This rules out feedback from y to future values of z .

Assumption TS.3 has the advantage of being more realistic about the random nature of the x_{it} , while it isolates the necessary assumption about how u_t and the explanatory variables are related in order for OLS to be unbiased.

Theorem 1 – Unbiasedness of OLS

Under Assumptions TS.1, TS.2, and TS.3, the OLS estimators are unbiased conditional on $\mathbf{X}$, and therefore unconditionally as well when the expectations exist:

$$\mathbb{E}(\hat{\beta}_j) = \beta_j, \quad j = 0, 1, \dots, k.$$

If this assumption fails, OLS is not guaranteed to be unbiased.

1.3.2 The Variances of the OLS Estimators and the Gauss-Markov Theorem

Assumption TS.4 – Homoskedasticity

Conditional on $\mathbf{X}$, the variance of u_t is the same for all t :

$$\text{Var}(u_t | \mathbf{X}) = \text{Var}(u_t) = \sigma^2, \quad t = 1, 2, \dots, n.$$

This assumption means that the variance of the error does not depend on the explanatory variables $\mathbf{X}$, and that the variance is constant over time. It is sufficient that u_t and $\mathbf{X}$ are independent.

Assumption TS.5 – No Serial Correlation

Conditional on $\mathbf{X}$, the errors in two different time periods are uncorrelated:

$$\text{Corr}(u_t, u_s | \mathbf{X}) = 0, \quad \text{for all } t \neq s.$$

The simplest way to understand TS.5 is that the errors u_t should be uncorrelated across time, after conditioning on the regressors $\mathbf{X}$. This means:

$$\text{Corr}(u_t, u_{t-j}) = 0, \quad \text{for all } j \neq 0. \quad (10)$$

When this fails, we say there is **serial correlation** or **autocorrelation**. That is, errors from adjacent time periods tend to move together.

Assumptions TS.1 through TS.5 are the time series versions of the Gauss-Markov assumptions. They ensure that:

- the OLS estimators are unbiased,
- we can calculate valid standard errors.

Theorem 10.2 – OLS Sampling Variances

Under the time series Gauss-Markov Assumptions TS.1 through TS.5, the variance of $\hat{\beta}_j$, conditional on $\mathbf{X}$, is

$$\text{Var}(\hat{\beta}_j | \mathbf{X}) = \frac{\sigma^2}{SST_j(1 - R_j^2)}, \quad j = 1, \dots, k, \quad (11)$$

where:

- SST_j is the total sum of squares of x_j ,
- R_j^2 is the R^2 from regressing x_j on the other independent variables.

Theorem 10.3 – Unbiased Estimation of σ^2

Under Assumptions TS.1 through TS.5, the estimator $\hat{\sigma}^2 = \frac{SSR}{n-k-1}$ is an unbiased estimator of σ^2 , where SSR is the sum of squared residuals. i.e $\mathbb{E}(\hat{\sigma}^2) = \sigma^2$.

Theorem 10.4 – Gauss–Markov Theorem

Under Assumptions TS.1 through TS.5, the OLS estimators are the **best linear unbiased estimators** (BLUE), conditional on $\mathbf{X}$.

This final result tells us that, among all linear and unbiased estimators, OLS achieves the smallest variance—just as in the cross-sectional setting.

1.3.3 Inference under the Classical Linear Model Assumptions

Assumption TS.6 – Normality

The errors u_t are independent of $\mathbf{X}$ and are independently and identically distributed as:

$$u_t \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(0, \sigma^2)$$

Assumption TS.6 implies TS.3, TS.4, and TS.5, but it is stronger because it includes both independence and normality.

Theorem 10.5 – Normal Sampling Distributions

Under Assumptions TS.1 through TS.6, the classical linear model (CLM) assumptions for time series hold. Then:

- The OLS estimators are normally distributed, conditional on $\mathbf{X}$.
- Under the null hypothesis, each t -statistic follows a t -distribution.
- Each F -statistic follows an F -distribution.

It implies that, when Assumptions TS.1 through TS.6 hold, everything we have learned about estimation and inference for cross-sectional regressions applies directly to time series regressions. Thus, t statistics can be used for testing statistical significance of individual explanatory variables, and F statistics can be used to test for joint significance.

2 Further Issues in Using OLS with Time Series Data

2.1 Stationary and Nonstationary Time Series

2.1.1 Stationary and Weakly Dependent Time Series

The notion of a **stationary process** has played an important role in the analysis of time series. A stationary time series process is one in which the probability distributions are stable over time. If we take any collection of random variables in the sequence and then shift that sequence ahead h time periods, the joint probability distribution must remain unchanged.

A simpler, more useful way of describing a stochastic process is to give the **moments** of the process, particularly the first and second moments, which are called the **mean**, **variance**, and **autocovariance** functions.

Mean The mean function $\mu(t)$ is defined by:

$$\mu(t) = \mathbb{E}[X(t)]$$

Variance The variance function $\sigma^2(t)$ is defined by:

$$\sigma^2(t) = \text{Var}[X(t)]$$

Autocovariance We must define the **autocovariance function** $\gamma(t_1, t_2)$, which is the covariance of $X(t_1)$ with $X(t_2)$, namely:

$$\gamma(t_1, t_2) = \mathbb{E}[(X(t_1) - \mu(t_1))(X(t_2) - \mu(t_2))]$$

Note: The variance function is a special case of the autocovariance function when $t_1 = t_2$.

Stationary Stochastic Process. The stochastic process $\{x_t : t = 1, 2, \dots\}$ is **stationary** if for every collection of time indices $1 \leq t_1 < t_2 < \dots < t_k$, the joint distribution of $(x_{t_1}, x_{t_2}, \dots, x_{t_k})$ is the same as the joint distribution of $(x_{t_1+h}, x_{t_2+h}, \dots, x_{t_k+h})$ for all integers $h \geq 1$.

One implication (by choosing $m = 1$ and $t_1 = 1$) is that x_t has the same distribution as x_1 for all $t = 2, 3, \dots$. In other words, the sequence $\{x_t : t = 1, 2, \dots\}$ is *identically distributed*. Again, this places no restrictions on how x_t and x_{t+1} are related to one another; indeed, they may be highly correlated.

A stochastic process that is not stationary is said to be a **nonstationary process**. It can be very difficult to determine whether the data we have collected were generated by a stationary process. However, it is easy to spot certain sequences that are not stationary.

Remark. In particular, if $n = 1$, strict stationarity implies that the distribution of $X(t)$ is the same for all t , so that, provided the first two moments are finite, we have

$$\mu(t) = \mu, \quad \sigma^2(t) = \sigma^2$$

which are both constants and do not depend on t .

If $n = 2$, the joint distribution of $X(t_1)$ and $X(t_2)$ depends only on $(t_2 - t_1)$, which is called the **lag**. Thus, the autocovariance function $\gamma(t_1, t_2)$ also depends only on $(t_2 - t_1)$, and may be written as $\gamma(\tau)$, where

$$\begin{aligned} \gamma(\tau) &= \mathbb{E}[(X(t) - \mu)(X(t + \tau) - \mu)] \\ &= \text{Cov}[X(t), X(t + \tau)] \end{aligned}$$

is called the **autocovariance coefficient at lag τ** and this will be abbreviated to acv.f.

The size of an autocovariance coefficient depends on the units in which $X(t)$ is measured. It's useful to standardize the acv.f. to produce a function called the **autocorrelation function**, defined as:

$$\rho(\tau) = \frac{\text{Cov}[X(t), X(t + \tau)]}{\sqrt{\text{Var}(X(t))}\sqrt{\text{Var}(X(t + \tau))}} = \frac{\gamma(\tau)}{\sqrt{\gamma(0)}\sqrt{\gamma(0)}} = \frac{\gamma(\tau)}{\gamma(0)}$$

Note: the variance of $X(t)$ is $\gamma(0)$, and by the stationarity, the the variance of $X(t)$ at any other time $X(t + \tau)$ is also $\gamma(0)$.

Covariance Stationary Process. A weaker and more practical condition is **covariance stationarity**. A process $\{x_t : t = 1, 2, \dots\}$ is covariance stationary if:

- $\mathbb{E}(x_t) = \mu$ is constant,
- $\text{Var}(x_t) = \sigma^2$ is constant,
- $\text{Cov}(x_t, x_{t+\tau}) = \gamma(\tau)$ depends only on the lag τ , not on the specific time t .
By letting $\tau = 0$, we note that the above assumption about the acv.f. implies that the variance, as well as the mean, is constant.

It only requires the first two moments and the autocovariances to be time-invariant.

2.1.2 Weakly Dependent Time Series

Loosely speaking, a stationary time series process $\{x_t : t = 1, 2, \dots\}$ is said to be **weakly dependent** if x_t and x_{t+h} are “almost independent” as $h \rightarrow \infty$, without bound.

Covariance stationary sequences can be characterized in terms of correlations: a covariance stationary time series is **weakly dependent** if the correlation between x_t and x_{t+h} goes to zero “sufficiently quickly” as $h \rightarrow \infty$. As the variables get farther apart in time, the correlation between them becomes smaller and smaller. Covariance stationary sequences where $\text{Corr}(x_t, x_{t+h}) \rightarrow 0$ as $h \rightarrow \infty$ are said to be **asymptotically uncorrelated**.

Remark. .

- It replaces the assumption of random sampling in implying that the LLN and the CLT hold.
- Stationary, weakly dependent time series are ideal for use in multiple regression analysis.
- A trending series, though certainly nonstationary, *can* be weakly dependent.

White Noise We use y to denote observed series of interest. Suppose that

$$y_t = e_t \quad e_t \sim (0, \sigma^2),$$

where the “shock,” e_t , is uncorrelated over time. We say that e_t , and hence y_t , is serially uncorrelated. We also assume that $\sigma^2 < \infty$. Such process, with zero mean, constant variance, and no serial correlation, is called zero-mean white noise, or simply white noise. We often write

$$e_t \sim \text{WN}(0, \sigma^2)$$

and hence

$$y_t \sim \text{WN}(0, \sigma^2).$$

Although ε_t , and hence y_t , are serially uncorrelated, they are not necessarily serially independent, because they are not necessarily normally distributed. If in addition to being serially uncorrelated, y is serially independent, then we say that y is **independent white noise**. We write:

$$y_t \stackrel{\text{iid}}{\sim} (0, \sigma^2)$$

and we say that “ y is independently and identically distributed with zero mean and constant variance.” Let’s characterize the dynamic stochastic structure of white noise, $y_t \sim \text{WN}(0, \sigma^2)$. By construction, the unconditional mean of y is

$$\mathbb{E}(y_t) = 0,$$

and the unconditional variance of y is

$$\text{Var}(y_t) = \sigma^2.$$

The unconditional mean and variance are constant. In fact, the unconditional mean and variance must be constant for any covariance stationary process.

We need to compute and examine its mean and its autocovariance function. Fortunately, however, it’s very simple. Because white noise is, by definition, uncorrelated over time, all the autocovariances, and hence all the autocorrelations, are zero beyond displacement 0. The autocovariance function for a white noise process is

$$\gamma(\tau) = \begin{cases} \sigma^2, & \tau = 0 \\ 0, & \tau \geq 1 \end{cases}$$

and the autocorrelation function for a white noise process is

$$\rho(\tau) = \begin{cases} 1, & \tau = 0 \\ 0, & \tau \geq 1. \end{cases}$$

Moving Average Process of Order “1” The simplest example of a weakly dependent time series is an independent, identically distributed sequence: a sequence that is independent is trivially weakly dependent. A more interesting example of a weakly dependent sequence is

$$x_t = e_t + \alpha_1 e_{t-1}, \quad t = 1, 2, \dots, \quad (1)$$

where $e_t \stackrel{iid}{\sim} (0, \sigma_e^2)$. The process $\{x_t\}$ is called a **moving average process of order one [MA(1)]**: x_t is a weighted average of e_t and e_{t-1} ; in the next period, we drop e_{t-1} , and then x_{t+1} depends on e_t and e_{t+1} . Setting the coefficient of e_t to 1 in (1) is done without loss of generality. Adjacent terms in the sequence are correlated:

$$\begin{aligned} \mathbb{E}(x_t) &= \mathbb{E}(e_t + \alpha_1 e_{t-1}) = 0 \\ \mathbb{E}(x_{t+h}) &= \mathbb{E}(e_{t+h} + \alpha_1 e_{t+h-1}) = 0 \\ \text{Var}(x_t) &= \text{Var}(e_t + \alpha_1 e_{t-1}) = \text{Var}(e_t) + \text{Var}(\alpha_1 e_{t-1}) = \sigma_e^2 + \alpha^2 \cdot \sigma_e^2 = (1 + \alpha^2) \sigma_e^2 \\ \text{Var}(x_{t+h}) &= \text{Var}(e_{t+h} + \alpha_1 e_{t+h-1}) = \text{Var}(e_{t+h}) + \text{Var}(\alpha_1 e_{t+h-1}) = \sigma_e^2 + \alpha^2 \cdot \sigma_e^2 = (1 + \alpha^2) \sigma_e^2 \\ \text{Cov}(x_t, x_{t+1}) &= \text{Cov}(e_t + \alpha_1 e_{t-1}, e_{t+1} + \alpha_1 e_t) = \text{Cov}(e_t, \alpha_1 e_t) = \alpha_1 \sigma_e^2 \quad (\text{by linearity}) \\ \text{Cov}(x_t, x_{t+2}) &= 0 \\ \text{Cov}(x_t, x_{t+h}) &= 0 \end{aligned}$$

We can conclude the autocovariance function to

$$\text{Cov}(x_t, x_{t+h}) = \gamma(h) = \begin{cases} \alpha_1 \sigma_e^2, & h = 1 \\ 0, & h > 1 \end{cases}$$

Thus, an MA(1) is a stationary, weakly dependent sequence, and the LLN and the CLT can be applied to $\{x_t\}$.

Autoregressive Process of Order “1” A more popular example is the process

$$y_t = \rho_1 y_{t-1} + e_t, \quad t = 1, 2, \dots \quad (2)$$

The starting point in the sequence is y_0 , and $\{e_t\}$ is an i.i.d. sequence with zero mean and variance σ_e^2 . Assume that y_0 is independent of the errors and that $\mathbb{E}(y_0) = 0$. This is called an **autoregressive process of order one (AR(1))**. The crucial assumption for weak dependence of an AR(1) process is the **stability condition**:

$$|\rho_1| < 1.$$

To see that a stable AR(1) process is asymptotically uncorrelated, assume it is covariance stationary. Then:

$$\mathbb{E}(y_t) = \mathbb{E}(y_{t-1}) = \mu$$

and from (2),

$$\mathbb{E}(y_t) = \rho_1 \mathbb{E}(y_{t-1}) + \mathbb{E}(e_t) \Rightarrow \mu = \rho_1 \mu \Rightarrow \mu(1 - \rho_1) = 0 \Rightarrow \mu = 0.$$

Now taking variance on both sides of (2):

$$\text{Var}(y_t) = \rho_1^2 \text{Var}(y_{t-1}) + \text{Var}(e_t) \Rightarrow \sigma_y^2 = \rho_1^2 \sigma_y^2 + \sigma_e^2 \Rightarrow \sigma_y^2 = \frac{\sigma_e^2}{1 - \rho_1^2}. \quad (3)$$

Now compute the covariance between y_t and y_{t+h} , for $h \geq 1$:

$$\begin{aligned} y_{t+h} &= \rho_1 y_{t+h-1} + e_{t+h} \\ &= \rho_1 (\rho_1 y_{t+h-2} + e_{t+h-1}) + e_{t+h} \\ &= \rho_1^2 y_{t+h-2} + \rho_1 e_{t+h-1} + e_{t+h} \\ &\vdots \\ &= \rho_1^h y_t + \rho_1^{h-1} e_{t+1} + \dots + e_{t+h} \end{aligned}$$

Because $\mathbb{E}(y_t) = 0$ for all t , we can multiply this last equation by y_t and take expectations to obtain $\text{Cov}(y_t, y_{t+h})$. Using the fact that e_{t+j} is uncorrelated with y_t for all $j \geq 1$ gives

$$\begin{aligned}\text{Cov}(y_t, y_{t+h}) &= \mathbb{E}(y_t y_{t+h}) \\ &= \rho_1^h \mathbb{E}(y_t^2) + \rho_1^{h-1} \mathbb{E}(y_t e_{t+1}) + \cdots + \mathbb{E}(y_t e_{t+h}) \\ &= \rho_1^h \mathbb{E}(y_t^2) = \rho_1^h \sigma_y^2.\end{aligned}$$

Because σ_y is the standard deviation of both y_t and y_{t+h} , and the correlation between y_t and y_{t+h} for any $h \geq 1$:

$$\text{Corr}(y_t, y_{t+h}) = \frac{\text{Cov}(y_t, y_{t+h})}{\sigma_y \sigma_y} = \rho_1^h. \quad (4)$$

$\text{Corr}(y_t, y_{t+1}) = \rho_1$, so ρ_1 is the correlation coefficient between any two adjacent terms in the sequence. Equation (4) is important because it shows that, although y_t and y_{t+h} are correlated for any $h \geq 1$, this correlation gets very small for large h : because $|\rho_1| < 1$, $\rho_1^h \rightarrow 0$ as $h \rightarrow \infty$.

2.2 Asymptotic Properties of OLS

Assumption TS.1' – Linearity in Parameters

We assume the model is exactly as in Assumption TS.1, but now we add the assumption that $\{(x_t, y_t) : t = 1, 2, \dots\}$ is **stationary** and **weakly dependent**. In particular, the LLN and the CLT can be applied to sample averages.

The linear in parameters requirement again means that we can write the model as:

$$y_t = \beta_0 + \beta_1 x_{t1} + \cdots + \beta_k x_{tk} + u_t, \quad (5)$$

where the β_j are the parameters to be estimated. Unlike in Chapter 10, the x_{tj} can include lags of the dependent variable.

The important extra restriction in Assumption TS.1' as compared with Assumption TS.1 is the weak dependence assumption. Technically, Assumption TS.1' requires weak dependence on multiple time series (y_t and elements of $\mathbf{x}_t$, and this entails putting restrictions on the joint distribution across time.

Assumption TS.2' – No Perfect Collinearity

Same as Assumption TS.2.

In the sample (and therefore in the underlying time series process), no independent variable is constant nor a perfect linear combination of the others.

Assumption TS.3' – Zero Conditional Mean

The explanatory variables $\mathbf{x}_t = (x_{t1}, x_{t2}, \dots, x_{tk})$ are **contemporaneously exogenous**:

$$\mathbb{E}(u_t \mid \mathbf{x}_t) = 0.$$

This is the most natural assumption concerning the relationship between u_t and the explanatory variables. It is much weaker than Assumption TS.3 because it puts no restrictions on how u_t is related to the explanatory variables in other time periods.

- By stationarity, if contemporaneous exogeneity holds for one time period, it holds for them all. Relaxing stationarity would simply require us to assume the condition holds for all $t = 1, 2, \dots$
- Consistency result only requires u_t to have zero unconditional mean and to be uncorrelated with each x_{tj} :

$$\mathbb{E}(u_t) = \mathbb{E}[\mathbb{E}(u_t \mid \mathbf{x}_t)] = 0, \quad \text{Cov}(x_{tj}, u_t) = 0, \quad j = 1, \dots, k. \quad (6)$$

Theorem 11.1 – Consistency of OLS

Under TS.1', TS.2', and TS.3', the OLS estimators are consistent:

$$\text{plim } \hat{\beta}_j = \beta_j, \quad j = 0, 1, \dots, k.$$

Some key differences between Thm 10.1 and Thm 11.1:

1. We conclude that the OLS estimators are consistent, but not necessarily unbiased.
2. Thm 11.1, we have weakened the sense in which the explanatory variables must be exogenous, but weak dependence is required in the underlying time series.

We examine by some examples

Example 11.1 – Static Model

Consider a static model with two explanatory variables:

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + u_t.$$

Under weak dependence, the condition sufficient for consistency of OLS is:

$$\mathbb{E}(u_t \mid x_{t1}, x_{t2}) = 0.$$

This rules out omitted variables that are in u_t and are correlated with either x_{t1} or x_{t2} . Also, no function of x_{t1} or x_{t2} can be correlated with u_t , and so Assumption TS.3' rules out misspecified functional form.

Importantly, Assumption TS.3' **does not** rule out correlation between, say, u_{t-1} and z_t . If z_t is related to past y_{t-1} , such as

$$z_t = \delta_0 + \delta_1 y_{t-1} + v_t. \quad (7)$$

Such a mechanism generally causes z_t and u_{t-1} to be correlated (as can be seen by plugging in for y_{t-1}). This kind of feedback is allowed under Assumption TS.3'.

The next example clearly violates the strict exogeneity assumption; therefore, we can only appeal to large sample properties of OLS.

Example 11.3 – AR(1) Model

Consider the AR(1) model,

$$y_t = \beta_0 + \beta_1 y_{t-1} + u_t, \quad (8)$$

where the error u_t has a zero expected value, given all past values of y :

$$\mathbb{E}(u_t \mid y_{t-1}, y_{t-2}, \dots) = 0. \quad (9)$$

Combined, these two equations imply that

$$\mathbb{E}(y_t \mid y_{t-1}, y_{t-2}, \dots) = \mathbb{E}(y_t \mid y_{t-1}) = \beta_0 + \beta_1 y_{t-1}. \quad (10)$$

This result is very important.

1. It means that, once y_{t-1} has been controlled for, no further lags of y affect the expected value of y_t .
2. The relationship is assumed to be linear.

Because $\mathbf{x}_t$ contains only y_{t-1} , equation (9) implies that Assumption TS.3' holds. By contrast, the strict exogeneity assumption needed for unbiasedness, Assumption TS.3, does not hold. In fact, because u_t is uncorrelated with y_{t-1} , under (9), u_t and y_t must be correlated. In fact, it is easily seen that $\text{Cov}(y_t, u_t) = \text{Var}(u_t) > 0$. Therefore, a model with a lagged dependent variable cannot satisfy the strict exogeneity Assumption TS.3.

For the weak dependence condition to hold, we must assume that $|\beta_1| < 1$. If this condition holds,

then Theorem 11.1 implies that the OLS estimator from the regression of y_t on y_{t-1} produces consistent estimators of β_0 and β_1 .

Assumption TS.4' – Homoskedasticity

The errors are **contemporaneously homoskedastic**, that is, $\text{Var}(u_t | \mathbf{x}_t) = \sigma^2$.

Assumption TS.5' – No Serial Correlation

For all $t \neq s$,

$$\mathbb{E}(u_t u_s | \mathbf{x}_t, \mathbf{x}_s) = 0.$$

In TS.4', note how we condition only on the explanatory variables at time t . In TS.5', we condition only on the explanatory variables in the time periods coinciding with u_t and u_s . Assumption TS.5' **does** hold in the AR(1) model stated in equations (11.12) and (11.13). Because the explanatory variable at time t is y_{t-1} , we must show that

$$\mathbb{E}(u_t u_s | y_{t-1}, y_{t-2}, \dots) = 0 \quad \text{for all } t \neq s.$$

Suppose that $s < t$. (The other case follows by symmetry.) Then, because

$$u_s = y_s - \beta_0 - \beta_1 y_{s-1},$$

it follows that u_s is a function of y dated before time t . But by (9),

$$\mathbb{E}(u_t | u_s, y_{t-1}, y_{s-1}) = 0.$$

Therefore,

$$\mathbb{E}(u_t u_s | u_s, y_{t-1}, y_{s-1}) = u_s \mathbb{E}(u_t | y_{t-1}, y_{s-1}) = 0.$$

By the law of iterated expectations,

$$\mathbb{E}(u_t u_s | y_{t-1}, y_{s-1}) = 0.$$

This is a very important result: as long as only one lag appears in the model (8), the errors must be serially uncorrelated.

Theorem 11.2 – Asymptotic Normality of OLS

Under TS.1' through TS.5', the OLS estimators are asymptotically normally distributed. Further, the usual OLS standard errors, t statistics, F statistics, and LM statistics are asymptotically valid.

Even if the classical linear model assumptions do not hold, OLS is still consistent, and the standard inference procedures are valid.

2.3 Using Highly Persistent Time Series in Regression Analysis

We provide some examples of **highly persistent** (or **strongly dependent**) time series and show how they can be transformed for use in regression analysis.

2.3.1 Highly Persistent Time Series

In the simple AR(1) model (2), the assumption $|\rho_1| < 1$ is crucial for the series to be weakly dependent. However, AR(1) model with $\rho_1 = 1$, in this case we write

$$y_t = y_{t-1} + e_t, \quad t = 1, 2, \dots \quad (11)$$

where we assume that $\{e_t : t = 1, 2, \dots\}$ is independent and identically distributed with mean zero and variance σ_e^2 also the initial value y_0 is independent of e_t for all $t \geq 1$.

Eq. (11) is called a **random walk**. The fact that y_t at time t is obtained by starting at the previous value, y_{t-1} , and adding a zero mean random variable that is independent of y_{t-1} .

We find the expected value of y_t . This is most easily done by using repeated substitution to get:

$$y_t = e_t + e_{t-1} + \cdots + e_1 + y_0 = y_0 + \sum_{i=1}^t e_i.$$

Taking the expected value of both sides gives:

$$\begin{aligned}\mathbb{E}(y_t) &= \mathbb{E}(e_t) + \mathbb{E}(e_{t-1}) + \cdots + \mathbb{E}(e_1) + \mathbb{E}(y_0) \\ &= \mathbb{E}(y_0), \quad \text{for all } t \geq 1.\end{aligned}$$

The expected value of a random walk *does not depend on* t . A popular assumption is that $y_0 = 0$ —the process begins at zero at time zero—in which case, $\mathbb{E}(y_t) = 0 \forall t$.

The variance of a random walk does change with t . Assuming e_t is iid with variance σ^2 and y_0 is nonrandom so that $\text{Var}(y_0) = 0$, we have:

$$\begin{aligned}\text{Var}(y_t) &= \text{Var}(e_t) + \text{Var}(e_{t-1}) + \cdots + \text{Var}(e_1) + \text{Var}(y_0) \\ &= \text{Var}\left(y_0 + \sum_{i=1}^t e_i\right) \\ &= t\sigma_e^2.\end{aligned}$$

In other words, the variance of a random walk increases as a linear function of time. This shows that the process **cannot be stationary**.

A random walk displays highly persistent behavior in the sense that the value of y_0 today is important for determining the value of y_t in the very distant future. We write for h periods hence,

$$y_{t+h} = e_{t+h} + e_{t+h-1} + \cdots + e_{t+1} + y_t = \sum_{i=t+1}^h e_i + y_t.$$

We check the variance first

$$\begin{aligned}\text{Var}(y_{t+h}) &= \text{Var}(e_{t+h}) + \text{Var}(e_{t+h-1}) + \cdots + \text{Var}(e_t) + \text{Var}(y_t) \\ &= \text{Var}\left(y_t + \sum_{i=t+1}^{t+h} e_i\right) \\ &= \text{Var}\left(y_0 + \sum_{i=1}^t e_i + \sum_{i=t+1}^{t+h} e_i\right) \\ &= (t+h)\sigma_e^2.\end{aligned}$$

and see the covariance between y_t, y_{t+h}

$$\text{Cov}(y_t, y_{t+h}) = \text{Cov}\left(y_0 + \sum_{i=1}^t e_i, y_0 + \sum_{i=1}^t e_i + \sum_{i=t+1}^{t+h} e_i\right) = t\sigma_e^2.$$

also the correlation is

$$\text{Corr}(y_t, y_{t+h}) = \frac{\text{Cov}(y_t, y_{t+h})}{\sqrt{\text{Var}(y_t)}\sqrt{\text{Var}(y_{t+h})}} = \frac{t\sigma_e^2}{\sqrt{t\sigma_e^2}\sqrt{(t+h)\sigma_e^2}} = \sqrt{\frac{t}{t+h}}$$

Remark. .

- A random walk is a special case of what is known as a **unit root process**.
- A more general class of unit root processes is generated as in (11), but $\{e_t\}$ is now allowed to be a general, weakly dependent series.

- When $\{e_t\}$ is not an i.i.d. sequence, the properties of the random walk we derived earlier no longer hold.
- Key feature of $\{y_t\}$ is preserved: the value of y today is highly correlated with y even in the distant future.
- It is often important to know whether an economic time series is highly persistent or not.

It is extremely important not to confuse trending and highly persistent behaviors. A series can be trending but not highly persistent. It is often the case that a highly persistent series also contains a clear trend. One model that leads to this behavior is the **random walk with drift**:

$$y_t = \alpha_0 + y_{t-1} + e_t, \quad t = 1, 2, \dots \quad (12)$$

where $\{e_t : t = 1, 2, \dots\}$ and y_0 satisfy the same properties as in the random walk model. The parameter α_0 , which is called the *drift term*. Generating y_t the constant α_0 is added along with the random noise e_t to the previous value y_{t-1} .

$$y_t = \alpha_0 t + e_t + e_{t-1} + \dots + e_1 + y_0.$$

Therefore, if $y_0 = 0$, $E(y_t) = \alpha_0 t$: the expected value of y_t is growing over time if $\alpha_0 > 0$ and shrinking over time if $\alpha_0 < 0$.

$$E(y_t) = E(y_0) + t\alpha_0$$

$$\text{Var}(y_t) = t\sigma_e^2$$

$$\text{Cov}(y_t, y_{t_h}) = t\sigma^2$$

Remarks on Forecasts Let Ω_t be the information set at time t , then

Model	Forecast: $E[y_{t+h} \mid \Omega_t]$ with Derivation
AR(1) $y_t = \alpha_0 + \rho y_{t-1} + \varepsilon_t$	$E[y_{t+h} \mid \Omega_t] = \rho^h y_t + \alpha_0 \sum_{j=0}^{h-1} \rho^j = \rho^h y_t + \alpha_0 \cdot \frac{1 - \rho^h}{1 - \rho}$ As $h \rightarrow \infty$, $\rho^h \rightarrow 0 \Rightarrow E[y_{t+h} \mid \Omega_t] \rightarrow \frac{\alpha_0}{1 - \rho} = E(y_t)$
MA(1) $y_t = \varepsilon_t + \beta_1 \varepsilon_{t-1}$	$\begin{cases} E[y_{t+1} \mid \Omega_t] = \beta_1 \varepsilon_t & (\text{since } \varepsilon_t \text{ is known}) \\ E[y_{t+h} \mid \Omega_t] = 0, & h > 1 \text{ (all terms are future shocks)} \end{cases}$
RandomWalk (RW) $y_t = y_{t-1} + \varepsilon_t$	$y_{t+h} = y_t + \sum_{j=1}^h \varepsilon_{t+j} \Rightarrow E[y_{t+h} \mid \Omega_t] = y_t$ (since all future shocks have mean 0)
RW with Drift $y_t = \alpha + y_{t-1} + \varepsilon_t$	$y_{t+h} = y_t + h\alpha + \sum_{j=1}^h \varepsilon_{t+j} \Rightarrow E[y_{t+h} \mid \Omega_t] = y_t + h\alpha$ (drift accumulates linearly over time)

2.3.2 Transformations on Highly Persistent Time Series

Weakly dependent processes are said to be **integrated of order zero**, or $I(0)$ which means averages of such sequences already satisfy the standard limit theorems. Unit root processes, such as a random walk (with or without drift), are said to be **integrated of order one**, or $I(1)$ whereas this means that the *first difference* of the process is weakly dependent (and often stationary). A time series that is $I(1)$ is often said to be a *difference-stationary process*. an $I(1)$ process is easiest to see for a random walk. With $\{y_t\}$ generated as in (11.20) for $t = 1, 2, \dots$,

$$\Delta y_t = y_t - y_{t-1} = e_t, \quad t = 2, 3, \dots \quad (13)$$

therefore, the first-differenced series $\{\Delta y_t : t = 2, 3, \dots\}$ is actually an i.i.d. sequence. More generally, if $\{y_t\}$ is generated by (13) where $\{e_t\}$ is any weakly dependent process, then $\{\Delta y_t\}$ is weakly dependent.

Remark. In actual data sets, if an original variable is named y , then its change or difference is often denoted cy or dy . For example, the change in *price* might be denoted cprice .)

Many time series y_t that are strictly positive are such that $\log(y_t)$ is integrated of order one. We can use the first difference in the logs, $\Delta \log(y_t) = \log(y_t) - \log(y_{t-1})$, in regression analysis. Because

$$\Delta \log(y_t) \approx \frac{(y_t - y_{t-1})}{y_{t-1}}, \quad (14)$$

Differencing time series before using them in regression analysis has another benefit: it removes any linear time trend. This is easily seen by writing a linearly trending variable as

$$y_t = \gamma_0 + \gamma_1 t + v_t,$$

where v_t has a zero mean. Then, $\Delta y_t = \gamma_1 + \Delta v_t$, and so $\mathbb{E}(\Delta y_t) = \gamma_1 + \mathbb{E}(\Delta v_t) = \gamma_1$. In other words, $\mathbb{E}(\Delta y_t)$ is constant. The same argument works for $\Delta \log(y_t)$ when $\log(y_t)$ follows a linear time trend.

2.3.3 Deciding Whether a Time Series Is $I(1)$

A very simple tool is motivated by the AR(1) model: if $|\rho_1| < 1$, then the process is $I(0)$, but it is $I(1)$ if $\rho_1 = 1$. When the AR(1) process is stable, $\rho_1 = \text{Corr}(y_t, y_{t-1})$. This sample correlation coefficient is called the **first order autocorrelation** of $\{y_t\}$; we denote this by $\hat{\rho}_1$. By applying the LLN, $\hat{\rho}_1$ can be shown to be consistent for ρ_1 *provided* $|\rho_1| < 1$.

3 Serial Correlation and Heteroskedasticity in Time Series Regressions

3.1 Properties of OLS with Serially Correlated Errors

3.1.1 Unbiasedness and Consistency of OLS with Serially Correlated Errors

- **Unbiasedness:** OLS estimators ($\hat{\beta}_j$) remain unbiased if the regressors are strictly exogenous (Assumption TS.3: $E(u_t|X) = 0$). This holds regardless of serial correlation in errors.
- **Consistency:** OLS estimators remain consistent under weaker contemporaneous exogeneity assumptions (e.g., $E(u_t|x_t) = 0$) and weak dependence of the time series, again, irrespective of serial correlation in errors.

3.1.2 Efficiency and Inference

- **Efficiency:** OLS is **no longer BLUE** (Best Linear Unbiased Estimator) in the presence of serial correlation, as Assumption TS.5 (no serial correlation, homoskedasticity) is violated.
- **Inference:** Usual OLS standard errors, t-statistics, and F-statistics are **not valid** (even asymptotically).

Derivation of $Var(\hat{\beta}_1)$ in Simple Regression with AR(1) Errors Model: $y_t = \beta_0 + \beta_1 x_t + u_t$, with $u_t = \rho u_{t-1} + e_t$, $|\rho| < 1$. Let $Var(u_t) = \sigma^2$ and $Cov(u_t, u_{t+j}) = \rho^j \sigma^2$. Assume $\bar{x} = 0$. The OLS estimator $\hat{\beta}_1 = \beta_1 + (\sum x_t u_t) / SST_x$. The variance of $\hat{\beta}_1$ is:

$$\begin{aligned} Var(\hat{\beta}_1 | X) &= \frac{1}{SST_x^2} Var\left(\sum_{t=1}^n x_t u_t \middle| X\right) \\ &= \frac{1}{SST_x^2} \left(\sum_{t=1}^n x_t^2 Var(u_t) + 2 \sum_{t=1}^{n-1} \sum_{j=1}^{n-t} x_t x_{t+j} Cov(u_t, u_{t+j}) \right) \\ &= \frac{\sigma^2 \sum x_t^2}{SST_x^2} + \frac{2\sigma^2}{SST_x^2} \sum_{t=1}^{n-1} \sum_{j=1}^{n-t} \rho^j x_t x_{t+j} \\ Var(\hat{\beta}_1 | X) &= \frac{\sigma^2}{SST_x} + \frac{2\sigma^2}{SST_x^2} \sum_{t=1}^{n-1} \sum_{j=1}^{n-t} \rho^j x_t x_{t+j} \quad [\text{Eq. 12.4 adapted}] \end{aligned}$$

The first term is the usual OLS variance. The second term is non-zero if $\rho \neq 0$.

- If $\rho > 0$ and x_t are positively serially correlated (common), the second term is positive.
- **Result:** The usual OLS variance formula (σ^2 / SST_x) typically *understates* the true variance. OLS standard errors are too small, and t-statistics are too large.

3.1.3 Goodness of Fit (R^2 and $\bar{R}^2$)

- R^2 and $\bar{R}^2$ remain **consistent estimators** of the population R-squared if the data are stationary and weakly dependent.
- This is analogous to the cross-sectional case with heteroskedasticity.
- Not meaningful for I(1) (non-stationary) processes.

3.1.4 Serial Correlation in the Presence of Lagged Dependent Variables

The statement “OLS is inconsistent with lagged DVs and serially correlated errors” is nuanced.

- **Case 1: OLS is Consistent.** If $y_t = \beta_0 + \beta_1 y_{t-1} + u_t$ and by construction $E(u_t | y_{t-1}, y_{t-2}, \dots) = 0$.
 - OLS is consistent for β_0, β_1 .

- However, u_t can still be serially correlated (e.g., if u_t is correlated with y_{t-2} , it will be correlated with u_{t-1}). Standard errors will be invalid.
- **Case 2: OLS is Inconsistent due to Misspecification.** If we specify $y_t = \beta_0 + \beta_1 y_{t-1} + u_t$ AND assume $u_t = \rho u_{t-1} + e_t$ where e_t is an innovation uncorrelated with past y 's.
 - Then y_{t-1} is correlated with u_t (because y_{t-1} is related to u_{t-1} , and u_t depends on u_{t-1}). Specifically, $Cov(y_{t-1}, u_t) = \rho Cov(y_{t-1}, u_{t-1}) \neq 0$.
 - This renders OLS inconsistent for estimating β_0, β_1 in the original equation.
 - **The true underlying model is likely misspecified.** Combining these equations shows y_t follows an AR(2) process:

$$y_t = \alpha_0 + \alpha_1 y_{t-1} + \alpha_2 y_{t-2} + e_t$$

OLS estimation of this correctly specified AR(2) model is consistent, and its error e_t is serially uncorrelated.

- **Bottom Line:** Serial correlation in errors of a dynamic model often implies the dynamic specification of the conditional mean is incomplete. The usual remedy is to include more lags of variables in the model rather than trying to “correct” for serial correlation in a misspecified dynamic model.

3.2 Serial Correlation–Robust Inference after OLS

- OLS estimators remain consistent under serial correlation (with appropriate exogeneity), but usual OLS standard errors and test statistics are invalid.
- This section details how to compute standard errors robust to serial correlation (and heteroskedasticity), often called HAC (Heteroskedasticity and Autocorrelation Consistent) standard errors.

3.2.1 General Approach and The Newey-West Estimator

Consider the model: $y_t = \beta_0 + \beta_1 x_{t1} + \dots + \beta_k x_{tk} + u_t$. We want a robust standard error for, say, $\hat{\beta}_1$.

Asymptotic Variance of $\hat{\beta}_1$ Let r_t be the (population) error from regressing x_{t1} on other x 's. The asymptotic variance of $\hat{\beta}_1$ is related to $Var(\sum_{t=1}^n r_t u_t)$. Let $a_t = r_t u_t$. The Newey-West method estimates $Var(\sum a_t)$.

Procedure for SC-Robust Standard Error for $\hat{\beta}_1$

1. **Initial OLS:** Estimate the main model y_t on $x_{t1}, \dots, x_{tk}$ by OLS. Obtain:
 - The OLS estimate $\hat{\beta}_1$.
 - The usual (potentially incorrect) OLS standard error: “ $se(\hat{\beta}_1)_{OLS}$ ”.
 - The standard error of the regression: $\hat{\sigma}$.
 - The OLS residuals: $\hat{u}_t$ for $t = 1, \dots, n$.
2. **Auxiliary Regression Residuals:** Regress x_{t1} on an intercept and $x_{t2}, \dots, x_{tk}$. Obtain the residuals $\hat{r}_t$.

$$x_{t1} = \delta_0 + \delta_2 x_{t2} + \dots + \delta_k x_{tk} + r_t \quad (1)$$

3. **Form $\hat{a}_t$:** For each t , calculate $\hat{a}_t = \hat{r}_t \hat{u}_t$.
4. **Compute $\hat{n}$:** For a chosen truncation lag $g > 0$, compute the Newey-West estimator for $Var(\sum a_t)$:

$$\hat{n} = \sum_{t=1}^n \hat{a}_t^2 + 2 \sum_{h=1}^g \left(1 - \frac{h}{g+1}\right) \sum_{t=h+1}^n \hat{a}_t \hat{a}_{t-h} \quad (2)$$

The term $(1 - \frac{h}{g+1})$ is the Bartlett kernel weight.

5. Compute SC-Robust (HAC) Standard Error for $\hat{\beta}_1$:

$$se(\hat{\beta}_1)_{HAC} = \left[\frac{se(\hat{\beta}_1)_{OLS}}{\hat{\sigma}} \right]^2 \sqrt{\hat{n}} \quad (3)$$

Alternatively, this is equivalent to $se(\hat{\beta}_1)_{HAC} = \frac{\sqrt{\hat{n}}}{\sum \hat{r}_t^2}$.

3.2.2 Choosing the Truncation Lag g

- g controls the number of autocovariances included.
- Theory suggests g should grow with sample size n , but at a slower rate (e.g., $g \propto n^{1/4}$ or $n^{1/3}$).
- Practical rules of thumb exist (e.g., EViews default $g \approx 4(n/100)^{2/9}$; Stock & Watson $g \approx (3/4)n^{1/3}$).
- Generally, small g (1 or 2) for annual data, larger for quarterly (e.g., 4, 8) or monthly (e.g., 12, 24) data, if n is sufficient.
- Results can be sensitive to the choice of g .

3.2.3 Properties and Use of HAC Standard Errors

- Robust to general forms of serial correlation (up to lag g) and heteroskedasticity.
- Typically larger than OLS SEs if errors are positively serially correlated.
- Can have poor performance in small samples if serial correlation is strong.
- Useful when strict exogeneity (for GLS) is doubtful, as OLS consistency is less demanding.

3.2.4 Kiefer and Vogelsang (2005) Alternative

- Proposes using a fixed fraction of the sample size for g , e.g., $g = n - 1$.
- The resulting t-statistic does not follow a standard normal distribution asymptotically.
- They provide alternative critical values, avoiding the need to select g based on growth rates relative to n .

3.2.5 SC-Robust F-statistics

- Possible to construct for multiple hypothesis tests, but more advanced.

3.3 Testing for Serial Correlation

- To inform the choice of appropriate standard errors (OLS vs. HAC).
- To decide if more efficient estimators like Feasible GLS (FGLS) are warranted (if strict exogeneity holds).
- As a model diagnostic, especially for dynamic models (potential for missing lags).

3.3.1 A t-Test for AR(1) Serial Correlation with Strictly Exogenous Regressors

Tests $H_0 : \rho = 0$ in $u_t = \rho u_{t-1} + e_t$.

1. Run OLS on $y_t = \beta_0 + \beta_1 x_{t1} + \dots + \beta_k x_{tk} + u_t$ to get residuals $\hat{u}_t$.
2. Run the auxiliary regression for $t = 2, \dots, n$:

$$\hat{u}_t = \hat{\rho} \hat{u}_{t-1} + \text{error}_t \quad (4)$$

(May include an intercept).

3. Use the t-statistic for $\hat{\rho}$, $t_{\hat{\rho}}$, to test H_0 .

Validity: Asymptotically valid if original regressors x_{tj} are strictly exogenous and innovations e_t in the AR(1) process for u_t are homoskedastic. Can be made robust to heteroskedastic e_t using robust SE for $\hat{\rho}$.

3.3.2 The Durbin-Watson (DW) Test under Classical Assumptions

Also tests $H_0 : \rho = 0$ in an AR(1) model for errors.

- Statistic: $DW = \frac{\sum_{t=2}^n (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^n \hat{u}_t^2}$.
- Approximation: $DW \approx 2(1 - \hat{\rho})$, where $\hat{\rho}$ is from Eq. (4).
- **Derivation of Approximation:**

$$\begin{aligned} DW &= \frac{\sum (\hat{u}_t^2 - 2\hat{u}_t\hat{u}_{t-1} + \hat{u}_{t-1}^2)}{\sum \hat{u}_t^2} \\ &\approx \frac{\sum \hat{u}_t^2 - 2\sum \hat{u}_t\hat{u}_{t-1} + \sum \hat{u}_{t-1}^2}{\sum \hat{u}_t^2} \quad (\text{since } \sum \hat{u}_{t-1}^2 \approx \sum \hat{u}_t^2) \\ &= 2 - 2\frac{\sum \hat{u}_t\hat{u}_{t-1}}{\sum \hat{u}_t^2} \approx 2(1 - \hat{\rho}) \end{aligned}$$

- Decision involves comparing DW to upper (d_U) and lower (d_L) critical value bounds. Test can be inconclusive.
- **Limitations:** Requires full CLM assumptions (including normality, strict exogeneity); inconclusive region; primarily for AR(1). Generally less preferred than the t-test.

3.3.3 Testing for AR(1) Serial Correlation without Strictly Exogenous Regressors

Crucial when regressors (e.g., lagged dependent variables) might be correlated with past errors.

1. Run OLS on the main model to get residuals $\hat{u}_t$.
2. Run the auxiliary regression for $t = 2, \dots, n$:

$$\hat{u}_t = \gamma_0 + \rho\hat{u}_{t-1} + \gamma_1x_{t1} + \dots + \gamma_kx_{tk} + \text{error}_t \quad (5)$$

(Includes original regressors x_{tj} from the main model).

3. Use the t-statistic for $\hat{\rho}$ from this regression to test $H_0 : \rho = 0$.

Validity: Asymptotically valid even if regressors are not strictly exogenous. Including x_{tj} controls for their potential correlation with $\hat{u}_{t-1}$. Assumes homoskedasticity of innovations e_t ; can use robust SE for $\hat{\rho}$.

3.3.4 Testing for Higher-Order Serial Correlation (e.g., AR(q))

Tests $H_0 : \rho_1 = \dots = \rho_q = 0$ in $u_t = \rho_1u_{t-1} + \dots + \rho_qu_{t-q} + e_t$.

1. Run OLS on the main model to get residuals $\hat{u}_t$.
2. Run the auxiliary regression for $t = q+1, \dots, n$:

$$\hat{u}_t = \gamma_0 + \rho_1\hat{u}_{t-1} + \dots + \rho_q\hat{u}_{t-q} + \gamma_1x_{t1} + \dots + \gamma_kx_{tk} + \text{error}_t \quad (6)$$

3. **F-test version:** Compute F-statistic for joint significance of $\hat{u}_{t-1}, \dots, \hat{u}_{t-q}$. Compare to $F_{q, n-(k+1)-q}$.
4. **LM-test version (Breusch-Godfrey):** Calculate $LM = (n-q)R_{\hat{u}}^2$, where $R_{\hat{u}}^2$ is from Eq. (6). Under H_0 , $LM \sim \chi_q^2$.

Validity: Valid with or without strict exogeneity of original x_{tj} (if included in auxiliary). Assumes homoskedastic e_t ; robust versions exist. If x_{tj} are strictly exogenous, they can be omitted from Eq. (6).

3.3.5 Testing for Seasonal Serial Correlation

- For data with seasonality (e.g., quarterly, monthly), test for correlation at seasonal lags.
- Example: For $u_t = \rho_s u_{t-s} + e_t$ (e.g., $s = 4$ for quarterly data), regress $\hat{u}_t$ on $\hat{u}_{t-s}$ (and x_{tj} if not strictly exogenous). Use t-statistic on coefficient of $\hat{u}_{t-s}$.

3.4 Correcting for Serial Correlation with Strictly Exogenous Regressors

3.4.1 Motivation and Pre-requisites

- **Goal:** Obtain more efficient estimators than OLS when errors are serially correlated, if OLS with HAC standard errors yields imprecise estimates.
- **Crucial Requirement:** Explanatory variables (X) must be **strictly exogenous** (error u_t uncorrelated with X in all time periods). Not suitable for models with lagged dependent variables.

3.4.2 Obtaining BLUE in AR(1) Model (Known ρ)

Assume errors u_t follow $u_t = \rho u_{t-1} + e_t$, with $|\rho| < 1$, and e_t are serially uncorrelated innovations with $Var(e_t) = \sigma_e^2$. Original model: $y_t = \beta_0 + X_t \beta + u_t$.

Transformation for $t \geq 2$ Subtracting $\rho \times$ (equation at $t - 1$) from (equation at t) yields:

$$(y_t - \rho y_{t-1}) = (1 - \rho)\beta_0 + (X_t - \rho X_{t-1})\beta + (u_t - \rho u_{t-1})$$

Let $\tilde{y}_t = y_t - \rho y_{t-1}$ and $\tilde{X}_t = X_t - \rho X_{t-1}$ (quasi-differenced data). The transformed equation is:

$$\tilde{y}_t = (1 - \rho)\beta_0 + \tilde{X}_t \beta + e_t, \quad \text{for } t = 2, \dots, n \quad (7)$$

The error e_t is serially uncorrelated with variance σ_e^2 .

Transformation for $t = 1$ (Prais-Winsten) The original error u_1 has $Var(u_1) = \sigma_e^2/(1 - \rho^2)$. To make its variance σ_e^2 , multiply the $t = 1$ equation by $\sqrt{1 - \rho^2}$:

$$\sqrt{1 - \rho^2} y_1 = \sqrt{1 - \rho^2} \beta_0 + (\sqrt{1 - \rho^2} X_1) \beta + \sqrt{1 - \rho^2} u_1$$

Let $\tilde{y}_1 = \sqrt{1 - \rho^2} y_1$, $\tilde{X}_1 = \sqrt{1 - \rho^2} X_1$. The transformed error $\sqrt{1 - \rho^2} u_1$ has variance σ_e^2 . **GLS Estimator:** OLS on the system of n transformed equations (Prais-Winsten for $t = 1$, quasi-difference for $t \geq 2$) is BLUE if ρ is known.

3.4.3 Feasible GLS (FGLS) Estimation with AR(1) Errors

Since ρ is unknown:

1. Estimate main model by OLS, get residuals $\hat{u}_t$.
2. Estimate ρ by regressing $\hat{u}_t$ on $\hat{u}_{t-1}$ to get $\hat{\rho}$.
3. Transform data using $\hat{\rho}$ (Prais-Winsten for $t = 1$, quasi-difference for $t \geq 2$).
4. Run OLS on the transformed data. The resulting $\hat{\beta}_{FGLS}$ are FGLS estimators.

$$\tilde{y}_t(\hat{\rho}) = \beta_0 \tilde{x}_{t0}(\hat{\rho}) + \tilde{X}_t(\hat{\rho}) \beta + \text{error}_t \quad (8)$$

(where $\tilde{x}_{t0}$ is the appropriately transformed intercept term).

5. Inference (t-stats, F-stats) from this final OLS is asymptotically valid.

Iterative Prais-Winsten/Cochrane-Orcutt: Repeat steps 2-4 until $\hat{\rho}$ converges. Asymptotically, one iteration is often enough.

3.4.4 Comparing OLS and FGLS

- OLS consistency requires $Cov(X_t, u_t) = 0$.
- FGLS consistency additionally requires (effectively) that u_t is uncorrelated with X_{t-1} and X_{t+1} .
- **Derivation of $E[(X_t - \rho X_{t-1})(u_t - \rho u_{t-1})] = 0$:** This requires $E[X_t u_t] - \rho E[X_t u_{t-1}] - \rho E[X_{t-1} u_t] + \rho^2 E[X_{t-1} u_{t-1}] = 0$. If $E[X_s u_s] = 0$, this simplifies to $-\rho(E[X_t u_{t-1}] + E[X_{t-1} u_t]) = 0$. Under stationarity, $E[X_t u_{t-1}] = E[X_{t+1} u_t]$, so we need $E[(X_{t+1} + X_{t-1})u_t] = 0$.
- If this stricter exogeneity for FGLS fails, OLS is consistent but FGLS is inconsistent. Significant differences between OLS and FGLS estimates might signal this problem.

3.4.5 Correcting for Higher-Order Serial Correlation (e.g., AR(2))

For $u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + e_t$:

- Transformation for $t \geq 3$:

$$(y_t - \rho_1 y_{t-1} - \rho_2 y_{t-2}) = \beta_0(1 - \rho_1 - \rho_2) + (X_t - \rho_1 X_{t-1} - \rho_2 X_{t-2})\beta + e_t$$

- FGLS: Estimate $\hat{\rho}_1, \hat{\rho}_2$ from OLS residuals ($\hat{u}_t$ on $\hat{u}_{t-1}, \hat{u}_{t-2}$). Transform data (complex for first two observations). Run OLS.

3.4.6 What if the Serial Correlation Model Is Wrong?

- FGLS estimators ($\hat{\beta}$) remain **consistent** if the regressors are strictly exogenous as required by FGLS, even if the assumed AR model (e.g., AR(1)) is incorrect.
- However, standard inference (t-stats, F-stats) from the FGLS transformed equation will be **invalid** because the transformed errors will still exhibit serial correlation or heteroskedasticity if the model was misspecified.
- **Robust Solution:** Apply Newey-West (HAC) standard errors to the OLS estimates obtained from the final FGLS transformed equation (e.g., eq. (8)). This makes inference robust to misspecification of the serial correlation structure and to heteroskedasticity in the underlying innovations.

3.5 Heteroskedasticity in Time Series Regressions

3.5.1 Heteroskedasticity-Robust Statistics

- OLS estimators remain unbiased/consistent with heteroskedasticity (given other assumptions hold).
- Usual OLS standard errors/tests are invalid.
- Heteroskedasticity-robust standard errors (e.g., White's) from Chapter 8 can be applied to time series if errors u_t are serially uncorrelated (Assumption TS.5' holds, only TS.4' - homoskedasticity - fails).

3.5.2 Testing for Heteroskedasticity

Tests like Breusch-Pagan (BP) or White can be used, but with caveats for time series:

1. Errors u_t from the main model should ideally not be serially correlated. Test for serial correlation first.
2. The error term (v_t) in the auxiliary regression for the test (e.g., $\hat{u}_t^2$ on x_{tj}) must be homoskedastic and serially uncorrelated for standard F/LM statistics to be valid.

BP Test Example:

1. OLS of y_t on X_t to get $\hat{u}_t$.
2. Regress $\hat{u}_t^2 = \delta_0 + \delta_1 x_{t1} + \dots + \delta_k x_{tk} + v_t$.
3. Test $H_0 : \delta_1 = \dots = \delta_k = 0$ (homoskedasticity).

3.5.3 Autoregressive Conditional Heteroskedasticity (ARCH)

A dynamic form of heteroskedasticity where $Var(u_t | \text{past } u)$ depends on past errors.

ARCH(1) Model Assume u_t are serially uncorrelated, i.e., $E(u_t | u_{t-1}, u_{t-2}, \dots) = 0$. The conditional variance is modeled as:

$$\sigma_t^2 \equiv Var(u_t | u_{t-1}, u_{t-2}, \dots) = E(u_t^2 | u_{t-1}) = \alpha_0 + \alpha_1 u_{t-1}^2 \quad (9)$$

- Requires $\alpha_0 > 0$ and $\alpha_1 \geq 0$ for non-negative variance.
- If $\alpha_1 = 0$, no ARCH effects.

We can write $u_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + v_t$, where $E(v_t | u_{t-1}, \dots) = 0$.

- **Stability Condition:** For a stable unconditional variance of u_t , we need $\alpha_1 < 1$.
- If $\alpha_1 > 0$, squared errors u_t^2 are serially correlated, even if u_t are not. This implies volatility clustering.

Implications of ARCH for OLS (when u_t are serially uncorrelated)

- If the original model $y_t = X_t\beta + u_t$ satisfies TS.1-TS.3 and TS.5 (serially uncorrelated u_t), OLS remains BLUE. ARCH violates TS.4 (homoskedasticity of u_t) in a specific way.
- More generally, if TS.1'-TS.5' hold (errors u_t are serially uncorrelated and satisfy $E(u_t | X_t) = 0$), OLS is consistent, and usual heteroskedasticity-robust standard errors (White's) are asymptotically valid as ARCH is a form of heteroskedasticity.
- **Why care about ARCH?**
 1. More efficient estimators of β may exist (e.g., WLS, MLE).
 2. Modeling conditional variance (volatility) is often of direct interest, especially in finance.

Testing for ARCH(1) Effects

1. Estimate the main model by OLS, get residuals $\hat{u}_t$.
2. Regress $\hat{u}_t^2$ on $\hat{u}_{t-1}^2$:

$$\hat{u}_t^2 = \hat{\alpha}_0 + \hat{\alpha}_1 \hat{u}_{t-1}^2 + \text{residual}_t$$

3. A statistically significant $\hat{\alpha}_1$ (via t-test) indicates ARCH(1) effects. (Can extend to ARCH(q) by including more lags and using an F-test).

3.5.4 Heteroskedasticity and Serial Correlation in Regression Models

When both are present:

- **Option 1: OLS with Fully Robust (HAC/Newey-West) Standard Errors.** This is a common and practical approach.
- **Option 2: Sequential Correction.**
 1. Test for serial correlation using a heteroskedasticity-robust test.
 2. If SC present, apply CO/PW transformation.
 3. Then, use heteroskedasticity-robust SEs on the FGLS estimates from the transformed equation, or test for heteroskedasticity in the (quasi-differenced) transformed equation's errors.
- **Option 3: Feasible GLS with Heteroskedasticity and AR(1) Serial Correlation.** Model: $y_t = X_t\beta + u_t$, where $u_t = \sqrt{h_t}\nu_t$ and $\nu_t = \rho\nu_{t-1} + e_t$. h_t is the heteroskedasticity function, e.g., $h_t = \exp(\delta_0 + X_t\delta)$. Procedure:
 1. OLS on main model, get $\hat{u}_t$.
 2. Estimate h_t : Regress $\log(\hat{u}_t^2)$ on X_t (or $\hat{y}_t, \hat{y}_t^2$), get fitted $\hat{g}_t$. Then $\hat{h}_t = \exp(\hat{g}_t)$.

3. Transform for heteroskedasticity: create $y_t^* = y_t/\sqrt{\hat{h}_t}$, $X_t^* = X_t/\sqrt{\hat{h}_t}$. The errors of $y_t^* = X_t^*\beta + \nu_t$ are ν_t , which are homoskedastic but AR(1).
4. Apply standard CO or PW methods to estimate the equation for y_t^* .

The resulting FGLS estimators can be asymptotically efficient. For robust inference, one could still apply HAC standard errors to the final estimates from step 4 if unsure about model specifications.

4 Advanced Time Series Topics

4.1 Testing for Unit Roots

4.1.1 Basic AR(1) Model for Unit Root Testing

The test typically starts with an AR(1) model for the time series y_t :

$$y_t = \alpha + \rho y_{t-1} + e_t, \quad t = 1, 2, \dots \quad (1)$$

where $E(e_t | y_{t-1}, y_{t-2}, \dots, y_0) = 0$. A unit root exists if $\rho = 1$.

4.1.2 Hypotheses and Transformed Equation

- **Null Hypothesis (H_0):** y_t has a unit root ($\rho = 1$).
- **Alternative Hypothesis (H_1):** y_t is stationary ($\rho < 1$).

A convenient form for testing is obtained by subtracting y_{t-1} from both sides of Eq. (1) and defining $\theta = \rho - 1$:

$$\Delta y_t = \alpha + \theta y_{t-1} + e_t \quad (2)$$

Now, $H_0 : \theta = 0$ versus $H_1 : \theta < 0$.

4.1.3 The Dickey-Fuller (DF) Test

- **Problem with Standard t-test:** Under $H_0 : \theta = 0$, y_{t-1} is I(1) (non-stationary). The t-statistic for $\hat{\theta}$ from OLS estimation of Eq. (2) does not follow a standard normal or t-distribution, even asymptotically. Its distribution is known as the **Dickey-Fuller distribution**.
- **DF Test Procedure:**
 1. Estimate Eq. (2) by OLS.
 2. Calculate the t-statistic $t_{\hat{\theta}} = \hat{\theta} / se(\hat{\theta})$.
 3. Compare $t_{\hat{\theta}}$ to critical values from the Dickey-Fuller distribution (e.g., Table 18.2 for no trend, Table 18.3 with trend).
- **Decision Rule:** Reject H_0 (unit root) if $t_{\hat{\theta}}$ is less than (more negative than) the appropriate Dickey-Fuller critical value.

Table 18.2 Asymptotic Critical Values (No Time Trend in test regression):

Significance level	1%	2.5%	5%	10%
Critical value	-3.43	-3.12	-2.86	-2.57

4.1.4 The Augmented Dickey-Fuller (ADF) Test

Used to account for serial correlation in Δy_t (which is e_t in Eq. (2) under H_0).

- **ADF Test Regression:** Add p lags of Δy_t to the DF regression:

$$\Delta y_t = \alpha + \theta y_{t-1} + \gamma_1 \Delta y_{t-1} + \dots + \gamma_p \Delta y_{t-p} + e_t \quad (3)$$

The goal is to make the error term e_t in this augmented regression serially uncorrelated.

- **Test Procedure:** Estimate Eq. (3) by OLS. The t-statistic for $\hat{\theta}$ on y_{t-1} is compared to the *same* Dickey-Fuller critical values as the basic DF test.
- Choosing p (number of lags) is important; too few leads to size distortion, too many reduces power. Often guided by data frequency or information criteria.

4.1.5 Testing for Unit Roots with a Time Trend

If y_t exhibits a trend, it might be trend-stationary (I(0) around a trend) rather than I(1).

- **Test Regression with Trend:** Include a deterministic time trend t in the ADF regression:

$$\Delta y_t = \alpha + \delta t + \theta y_{t-1} + \sum_{j=1}^p \gamma_j \Delta y_{t-j} + e_t \quad (4)$$

- **Critical Values:** The t-statistic for $\hat{\theta}$ is compared to *different, more negative* Dickey-Fuller critical values that account for the presence of the trend (e.g., Table 18.3).

Table 18.3 Asymptotic Critical Values (Linear Time Trend in test regression):

Significance level	1%	2.5%	5%	10%
Critical value	-3.96	-3.66	-3.41	-3.12

- The t-statistic on the trend coefficient $\hat{\delta}$ does not follow a standard distribution under $H_0 : \theta = 0$.

4.1.6 Important Considerations

- Failure to reject a unit root does not mean we "accept" it; it just means insufficient evidence against it. Tests can have low power.
- Tests are sensitive to sample size (low power in small samples) and structural breaks (bias towards not rejecting unit root).
- Other tests exist (e.g., Phillips-Perron, KPSS with $H_0 : I(0)$).

4.2 Spurious Regression

4.2.1 Definition and Context

- **Spurious Regression (Time Series with I(1) Variables):** Occurs when regressing one I(1) (unit root) time series on another *independent* I(1) time series.
- Such regressions often produce misleadingly "good" results:
 - High R^2 values.
 - Statistically significant t-statistics for slope coefficients.

Even though there is no true underlying relationship between the series.

- This is distinct from spurious correlation due to omitted common deterministic trends in I(0) series.

4.2.2 The Problem Illustrated: Independent Random Walks

Consider two independent random walks (I(1) processes):

$$x_t = x_{t-1} + a_t \quad (5)$$

$$y_t = y_{t-1} + e_t \quad (6)$$

where $\{a_t\}$ and $\{e_t\}$ are independent i.i.d. innovation processes with zero mean. This implies $\{x_t\}$ and $\{y_t\}$ are independent processes.

If we run the OLS regression:

$$y_t = \beta_0 + \beta_1 x_t + u_t \quad (7)$$

4.2.3 Why Standard OLS Fails (Consequences of I(1) Data)

- **Nature of the Error Term under H_0 :** If the true $\beta_1 = 0$ (no relationship), then $y_t = \beta_0 + u_t$. Since y_t is I(1), the error term u_t must also be I(1) (it inherits the random walk properties of y_t).
- **Violation of OLS Assumptions:** An I(1) error term u_t is:

- Non-stationary: Its variance grows with time.
- Highly serially correlated.

This violates even the asymptotic Gauss-Markov assumptions required for valid OLS inference (Chapter 11).

- **Non-Standard Asymptotic Distributions (Granger & Newbold, 1974):**

- The OLS estimator $\hat{\beta}_1$ does not converge to 0 in probability as would be expected for independent series.
- The **t-statistic for $\hat{\beta}_1$** does not follow a standard normal or t-distribution. It tends to be too large, leading to spurious rejection of $H_0 : \beta_1 = 0$ much more often than the nominal significance level suggests.
- The R^2 from the regression does not converge to a population constant. It converges to a non-degenerate random variable, meaning there's a high probability of observing a substantial R^2 even for independent I(1) series.

4.2.4 Implications and Further Points

- **Including a Time Trend:** Adding a deterministic time trend to the regression (7) does **not** solve the spurious regression problem if x_t and y_t are I(1) (e.g., random walks with drift). Detrending an I(1) series still leaves an I(1) series.
- **Multiple Regressors:** The problem extends to multiple regression if the dependent variable and at least one independent variable are I(1) and not cointegrated.
- **General I(0) Innovations:** The qualitative results hold even if a_t and e_t are general stationary I(0) processes, not just i.i.d.

4.2.5 Caution and Solutions

- Be extremely wary of regressing I(1) variables in their levels unless there's a strong theoretical basis for a long-run (cointegrating) relationship.
- **Differencing:** A common solution is to first-difference I(1) variables to achieve stationarity (Δy_t and Δx_t will be I(0)). Regressing Δy_t on Δx_t uses standard OLS inference if the differenced errors are well-behaved.
- **Cointegration (Topic of Section 18-4):** If a linear combination of I(1) variables is I(0), they are said to be cointegrated. In this specific case, regressions in levels can be meaningful.

4.3 Cointegration and Error Correction Models

4.4 Cointegration

If y_t and x_t are both I(1), they are **cointegrated** if a linear combination $s_t = y_t - \beta x_t$ is I(0) (stationary) for some cointegrating parameter β .

- This implies a **long-run equilibrium relationship**: $y_t \approx \beta x_t$.
- s_t is the deviation from equilibrium, which is temporary as it's I(0).

4.4.1 Testing for Cointegration

H_0 : No cointegration vs. H_1 : Cointegration.

1. **If β is known:** Form $s_t = y_t - \beta x_t$. Perform a DF/ADF test on s_t . Reject unit root in $s_t \implies$ cointegration.
2. **If β is unknown (Engle-Granger Test):**
 - (a) Estimate the cointegrating regression by OLS: $y_t = \hat{\alpha} + \hat{\beta}x_t + \hat{u}_t$. (May include constant/trend).
 - (b) Obtain residuals $\hat{u}_t$.

- (c) Perform a DF/ADF test on $\hat{u}_t$: Regress $\Delta\hat{u}_t = \theta\hat{u}_{t-1} + \text{lags of } \Delta\hat{u}_t + \text{error}$.
- (d) Compare the t-statistic of $\hat{\theta}$ to *special critical values* (e.g., Tables 18.4, 18.5), which are more negative than standard DF critical values.

Table 18.4 Critical Values (Cointegration Test, Levels reg. has constant):

Significance	1%	2.5%	5%	10%
Critical Value	-3.90	-3.59	-3.34	-3.04

Table 18.5 Critical Values (Cointegration Test, Levels reg. has constant & trend):

Significance	1%	2.5%	5%	10%
Critical Value	-4.32	-4.03	-3.78	-3.50

4.4.2 Estimating and Testing the Cointegrating Parameter β

- OLS estimator $\hat{\beta}$ from $y_t = \alpha + \beta x_t + u_t$ is consistent (super-consistent) if cointegrated.
- Standard inference on this $\hat{\beta}$ is problematic due to $I(1)$ x_t and potential endogeneity/serial correlation in u_t .
- **Leads and Lags Estimator:** To perform valid inference on β , augment the cointegrating regression:

$$y_t = \alpha_0 + \beta x_t + \phi_0 \Delta x_t + \sum_{j=1}^p \phi_j \Delta x_{t-j} + \sum_{j=1}^q \psi_j \Delta x_{t+j} + e_t$$

OLS on this equation yields $\hat{\beta}$. Standard t-tests (with HAC SEs for e_t if needed) are asymptotically valid for this $\hat{\beta}$.

4.4.3 Error Correction Models (ECM)

If y_t and x_t are cointegrated ($s_t = y_t - \beta x_t$ is $I(0)$), an ECM can model short-run dynamics and adjustment to long-run equilibrium.

- **Error Correction Term (ECT):** $s_{t-1} = y_{t-1} - \beta x_{t-1}$ (lagged deviation from equilibrium).
- **General ECM form:**

$$\Delta y_t = \alpha_0 + \sum \alpha_j \Delta y_{t-j} + \sum \gamma_j \Delta x_{t-j} + \delta s_{t-1} + u_t \quad (8)$$

All terms in this regression are $I(0)$.

- **Coefficient δ :** Speed of adjustment. Expected $\delta < 0$.
 - If $s_{t-1} > 0$ (y was above equilibrium), δs_{t-1} pulls Δy_t down.
 - If $s_{t-1} < 0$ (y was below equilibrium), δs_{t-1} pushes Δy_t up.
- **Estimation:**
 1. **If β known:** Use $s_{t-1} = y_{t-1} - \beta x_{t-1}$ directly in OLS estimation of Eq. (8).
 2. **If β unknown (Engle-Granger Two-Step):**
 - (a) Estimate β from the cointegrating regression $y_t = \alpha + \beta x_t + \text{error}$ to get $\hat{\beta}$.
 - (b) Form residuals $\hat{s}_{t-1} = y_{t-1} - \hat{\beta} x_{t-1}$.
 - (c) Estimate Eq. (8) by OLS, using $\hat{s}_{t-1}$ as the ECT.

Asymptotically, using $\hat{\beta}$ instead of true β does not affect inference on other ECM parameters ($\alpha_j, \gamma_j, \delta$).

4.5 Forecasting

4.5.1 Vector Autoregressive (VAR) Models

VAR models are used to capture the linear interdependencies among multiple time series. Each variable is modeled as a function of its own past values and the past values of all other variables in the system.

Structure of a VAR(p) Model For a bivariate system with variables y_t and z_t , a VAR(p) model is:

$$y_t = \alpha_1 + \sum_{i=1}^p \beta_{1i} y_{t-i} + \sum_{i=1}^p \gamma_{1i} z_{t-i} + u_{1t} \quad (9)$$

$$z_t = \alpha_2 + \sum_{i=1}^p \beta_{2i} y_{t-i} + \sum_{i=1}^p \gamma_{2i} z_{t-i} + u_{2t} \quad (10)$$

- p is the order of the VAR (number of lags).
- α_1, α_2 are intercepts.
- β_{ki}, γ_{ki} are coefficients for the lagged variables.
- u_{1t}, u_{2t} are innovation terms (shocks) for each equation.
 - Assumed serially uncorrelated: $E(u_{kt}u_{k,t-s}) = 0$ for $s \neq 0$.
 - Uncorrelated with past values of all variables.
 - Can be contemporaneously correlated: $E(u_{1t}u_{2t}) \neq 0$.
- Typically applied to stationary time series (I(0)). If variables are I(1), they are often differenced first, or a Vector Error Correction Model (VECM) is used if they are cointegrated.

Estimation and Lag Length Selection

- **Estimation:** Since each equation has the same regressors (lags of all variables), Ordinary Least Squares (OLS) applied equation by equation is consistent and asymptotically efficient.
- **Lag Length (p) Selection:** Crucial for model performance.
 - Methods include sequential F-tests or model selection criteria like AIC (Akaike Information Criterion) or SIC/BIC (Schwarz/Bayesian Information Criterion).

4.5.2 Granger Causality

Granger causality is a statistical concept indicating whether past values of one time series are useful for predicting another time series, beyond the information contained in the latter's own past values.

Concept

- Variable z_t **Granger-causes** variable y_t if knowledge of the past history of z_t significantly improves the prediction of y_t , given the past history of y_t .
- This is a test of *predictive causality*, not necessarily structural or economic causality.

Testing for Granger Causality using VAR

Does z Granger-cause y ? Based on equation (9) for y_t :

- **Null Hypothesis (H_0):** z does not Granger-cause y .

$$H_0 : \gamma_{11} = \gamma_{12} = \dots = \gamma_{1p} = 0$$

- **Alternative Hypothesis (H_1):** z Granger-causes y (at least one $\gamma_{1i} \neq 0$).
- **Test Procedure:**

1. **Unrestricted Model:** Estimate equation (9) by OLS. Obtain SSR_{UR} .
2. **Restricted Model:** Estimate equation (9) imposing H_0 (i.e., omitting all lags of z_t):

$$y_t = \alpha_1 + \sum_{i=1}^p \beta_{1i} y_{t-i} + u_{1t}^R$$

Obtain SSR_R .

3. **F-statistic:**

$$F = \frac{(SSR_R - SSR_{UR})/p}{SSR_{UR}/(n - k_{UR} - 1)}$$

where p is the number of restrictions (number of γ_{1i} coefficients), n is the number of observations used, and k_{UR} is the number of regressors in the unrestricted model (here, $k_{UR} = 2p$, excluding the intercept). So the denominator df is $n - 2p - 1$.

4. **Decision:** Reject H_0 if $F > F_{critical}$ from the $F(p, n - 2p - 1)$ distribution.

Does y Granger-cause z ? A symmetric procedure is applied to equation (10) for z_t , testing the null hypothesis $H_0 : \beta_{21} = \beta_{22} = \dots = \beta_{2p} = 0$.

Possible Outcomes:

- Unidirectional Granger causality (e.g., $z \rightarrow y$ only).
- Unidirectional Granger causality (e.g., $y \rightarrow z$ only).
- Bidirectional Granger causality (feedback: $z \leftrightarrow y$).
- No Granger causality in either direction.

Extension to More Variables: If more variables are in the VAR system, the test for z Granger-causing y involves testing the joint significance of all lags of z in the equation for y , which also includes lags of y and lags of all other system variables. The F-statistic construction remains similar, adjusting for the number of regressors.

5 Pooling Cross Sections across Time: Simple Panel Data Methods

5.1 Pooling Independent Cross Sections across Time

Definition and Purpose:

- **Independently Pooled Cross Section:** Formed by combining random samples drawn from a large population at *different points in time* (e.g., different years). Observations are independent within each cross-section and across time periods.
- **Key Reasons for Pooling:**
 - **Increase Sample Size:** Leads to more precise estimators (smaller standard errors) and more powerful statistical tests. Most effective if the underlying relationships (slopes) are stable over time.
 - **Analyze Changes Over Time:** Allows for studying how population characteristics, intercepts, or the effects (slopes) of variables change.

Econometric Modeling: The primary tool is Ordinary Least Squares (OLS) on the combined dataset, with modifications to account for time:

- Achieved by including **year dummy variables** for each time period except one (the base period).
- Model: $y_{it} = \beta_0 + \delta_1 d2_t + \delta_2 d3_t + \dots + \delta_{T-1} dT_t + \beta_1 x_{it1} + \dots + \beta_k x_{itk} + u_{it}$
- β_0 is the intercept for the base period.
- δ_j is the difference in the intercept for period $j + 1$ relative to the base period, *ceteris paribus*.
- Example 13.1 (Fertility): Shows how year dummy coefficients can reveal trends over time.

Allowing Slopes to Change Over Time

- Achieved by including **interaction terms** between year dummy variables and the explanatory variable(s) of interest.
- Model Example (for two periods, base and year “Y2”):

$$y_{it} = \beta_0 + \gamma_0 dY2_t + \beta_1 x_{it} + \gamma_1 (dY2_t \cdot x_{it}) + \text{other controls} + u_{it}$$

- β_1 = effect of x in the base period.
- $\beta_1 + \gamma_1$ = effect of x in year “Y2”.
- γ_1 = change in the effect of x from the base period to year “Y2”. A t-test on $\hat{\gamma}_1$ checks if this change is significant.
- Example 13.2 (Return to Education and Gender Wage Gap): Demonstrates testing for changes in slopes over time.

Important Statistical Considerations

- **Nominal vs. Real Data:**
 - For monetary variables in **logarithmic form**: Including year dummies often adequately controls for general inflation (affecting intercepts).
 - For monetary variables in **level form**: Essential to use **real (deflated) values** and include year dummies.
- **Heteroskedasticity:**
 - The error variance ($Var(u_{it})$) may differ across time or with other regressors.
 - Use **heteroskedasticity-robust standard errors** and test statistics.
- **Chow Test for Structural Change Across Time:**

- An F-test to determine if the entire regression function (intercepts and slopes) differs across time periods.
- Can be implemented by comparing Sums of Squared Residuals (SSRs) from pooled vs. separate regressions (not robust to heteroskedasticity).
- Alternatively, and more flexibly, by testing the joint significance of a year dummy and its interactions with all other explanatory variables (can be made robust).

5.2 Policy Analysis with Pooled Cross Sections

5.2.1 Introduction: Natural Experiments

- Pooled cross sections are valuable for evaluating policies/events via **natural experiments** (or quasi-experiments).
- These occur when an exogenous event (e.g., policy change) affects a **treatment group (T)** but not a **control group (C)**.
- Data is typically required from *before* and *after* the event/policy.

5.2.2 The Difference-in-Differences (DD or DID) Estimator

Setup and Model

- Requires two groups (T, C) and two time periods (1: Pre-policy, 2: Post-policy).
- Regression model estimated by OLS on pooled data:

$$y_{it} = \beta_0 + \delta_0 d2_t + \beta_1 dT_i + \delta_1 (d2_t \cdot dT_i) + \text{other factors} + u_{it}$$

- y_{it} : Outcome for unit i in period t .
- $d2_t$: Dummy for Period 2 (1 if post-policy, 0 if pre-policy). δ_0 = time trend for control group.
- dT_i : Dummy for Treatment group (1 if T, 0 if C). β_1 = pre-policy difference (T vs. C).
- $(d2_t \cdot dT_i)$: Interaction term. δ_1 is the DD estimate of the policy effect.

Derivation and Interpretation of $\hat{\delta}_1$ The OLS estimate $\hat{\delta}_1$ represents the policy effect.

- Expected values $E(y)$ (assuming $E(u_{it}) = 0$):
 - Control, Pre: β_0
 - Control, Post: $\beta_0 + \delta_0$
 - Treatment, Pre: $\beta_0 + \beta_1$
 - Treatment, Post: $\beta_0 + \delta_0 + \beta_1 + \delta_1$
- Change for Treatment Group ($\Delta E(y)_T$): $(\beta_0 + \delta_0 + \beta_1 + \delta_1) - (\beta_0 + \beta_1) = \delta_0 + \delta_1$
- Change for Control Group ($\Delta E(y)_C$): $(\beta_0 + \delta_0) - (\beta_0) = \delta_0$
- DD estimator: $\hat{\delta}_1 = \Delta E(y)_T - \Delta E(y)_C = (\delta_0 + \delta_1) - \delta_0 = (\bar{y}_{\text{Post},T} - \bar{y}_{\text{Post},C}) - (\bar{y}_{\text{Pre},T} - \bar{y}_{\text{Pre},C})$.
- Average Treatment effect (ATE): $\hat{\delta}_1 = (\bar{y}_{\text{Post},T} - \bar{y}_{\text{Pre},T}) - (\bar{y}_{\text{Post},C} - \bar{y}_{\text{Pre},C})$ which provides a different interpretation of the DD estimator. The first term, $(\bar{y}_{\text{Post},T} - \bar{y}_{\text{Pre},T})$, is the difference in means over time for the treated group. This would be a good estimator of the policy effect only if we can assume no external factors changed across the two time periods. While the second term, $(\bar{y}_{\text{Post},C} - \bar{y}_{\text{Pre},C})$ compute the same trend in averages for the control group.

We can estimate the δ_1 in two ways: (1) Compute the differences in averages between the treatment and control groups in each time period, and then difference the results over time; (2) Compute the change in averages over time for each of the treatment and control groups, and then difference these changes.

Parallel Trends Assumption

- **Crucial Assumption:** In the absence of the treatment, the average outcome for the treatment group would have followed the same trend as the average outcome for the control group.
- Violations lead to biased DD estimates.

Role of Other Explanatory Variables

- Including controls can account for compositional changes and improve precision of $\hat{\delta}_1$.

5.2.3 The Difference-in-Difference-in-Differences (DDD) Estimator

Motivation and Setup

- Used when the parallel trends assumption between the initial T and C groups is suspect.
- Introduces an additional layer of control, e.g., data from another state (State A, no policy) for the same groups (L: Low-income/target, M: Middle-income/control) as in the policy state (State B).

DDD Regression Model

$$y = \beta_0 + \beta_1 dL + \beta_2 dB + \beta_3(dL \cdot dB) + \delta_0 d2 + \delta_1(d2 \cdot dL) + \delta_2(d2 \cdot dB) + \delta_3(d2 \cdot dL \cdot dB) + u$$

- dL_i : Dummy for low-income group.
- dB_s : Dummy for State B (policy state).
- δ_3 on the triple interaction $d2_t \cdot dL_i \cdot dB_s$ is the DDD estimate.

Derivation and Interpretation of $\hat{\delta}_3$

- Let DD_B be the DD estimate in State B (policy state, comparing L and M groups): $DD_B = \delta_1 + \delta_3$.
- Let DD_A be the DD estimate in State A (control state, comparing L and M groups): $DD_A = \delta_1$.
- DDD estimator: $\delta_3 = DD_B - DD_A = (\delta_1 + \delta_3) - \delta_1$.
- $\hat{\delta}_3$ aims to net out common differential trends between L and M groups.

5.2.4 A General Framework for Policy Analysis with Pooled Cross Sections

Model for Complex Policy Rollouts

$$y_{igt} = \lambda_t + \alpha_g + \beta x_{gt} + \mathbf{z}_{igt}\gamma + u_{igt}$$

- y_{igt} : Outcome for unit i in group g at time t .
- λ_t : Time fixed effects (e.g., year dummies) for common time trends.
- α_g : Group fixed effects (e.g., state dummies) for time-invariant group differences.
- x_{gt} : Policy variable(s) for group g at time t . β is the effect of interest. It is important to not think that x_{gt} can always be constructed by interacting dummy variables indicating groups and time periods, as in the basic DD setup.
- $\mathbf{z}_{igt}$: Other control variables.

Estimation and Extensions

- Estimated by Pooled OLS with dummies for time periods and groups. Robust standard errors recommended.
- Can handle staggered policy adoption, multiple policies, continuous policies, and lagged effects.
- Can be extended to include group-specific time trends (e.g., $\alpha_g + c_g \cdot t$) if $T \geq 3$.
- Caution: Avoid perfect collinearity if x_{gt} only varies at the (g, t) level and full (g, t) interaction dummies are used.

5.3 Two-Period Panel Data Analysis

- **Panel Data (Longitudinal Data):** Involves observing the *same* cross-sectional units (e.g., individuals, firms, cities) over multiple time periods. This section focuses on the simplest case: two time periods ($t = 1$ and $t = 2$).
- Allows controlling for unobserved unit-specific characteristics that are constant over time.

5.3.1 The Unobserved Effects Model (Fixed Effects Model)

For unit i at time t , with a single explanatory variable x_{it} :

$$y_{it} = \beta_0 + \delta_0 d2_t + \beta_1 x_{it} + a_i + u_{it} \quad \text{for } t = 1, 2 \quad (1)$$

- y_{it} : Dependent variable.
- β_0 : Intercept for period 1.
- $d2_t$: Time dummy for period 2 ($d2_t = 1$ if $t = 2$, 0 if $t = 1$).
- δ_0 : Change in intercept from period 1 to period 2 (so intercept in period 2 is $\beta_0 + \delta_0$).
- x_{it} : Explanatory variable.
- β_1 : Slope coefficient of interest.
- a_i : **Unobserved effect** (or fixed effect / unobserved heterogeneity). Captures time-constant, unit-specific unobserved factors.
- u_{it} : **Idiosyncratic error** (or time-varying error). Captures time-varying unobserved factors.

5.3.2 The Problem with Pooled OLS

If we pool data from both periods and estimate $y_{it} = \beta_0 + \delta_0 d2_t + \beta_1 x_{it} + v_{it}$, where $v_{it} = a_i + u_{it}$ is the composite error:

- If a_i is correlated with x_{it} (i.e., $Cov(x_{it}, a_i) \neq 0$), then x_{it} is correlated with v_{it} .
- This leads to **biased and inconsistent** OLS estimators (**heterogeneity bias**).
- Serial correlation in v_{it} for the same unit i ($Cov(v_{i1}, v_{i2}) = Var(a_i)$) also violates OLS assumptions for standard error validity, even if $Cov(x_{it}, a_i) = 0$.

5.3.3 The First-Differencing (FD) Method

Motivation To eliminate the time-constant unobserved effect a_i , especially when it's correlated with x_{it} .

Derivation of the First-Differenced Equation

1. Model for Period 2 ($t = 2, d2_t = 1$):

$$y_{i2} = (\beta_0 + \delta_0) + \beta_1 x_{i2} + a_i + u_{i2}$$

2. Model for Period 1 ($t = 1, d2_t = 0$):

$$y_{i1} = \beta_0 + \beta_1 x_{i1} + a_i + u_{i1}$$

3. Subtracting Period 1 equation from Period 2 equation:

$$\begin{aligned} (y_{i2} - y_{i1}) &= ((\beta_0 + \delta_0) - \beta_0) + (\beta_1 x_{i2} - \beta_1 x_{i1}) + (a_i - a_i) + (u_{i2} - u_{i1}) \\ \Delta y_i &= \delta_0 + \beta_1 \Delta x_i + \Delta u_i \end{aligned} \quad (2)$$

where $\Delta y_i = y_{i2} - y_{i1}$, $\Delta x_i = x_{i2} - x_{i1}$, and $\Delta u_i = u_{i2} - u_{i1}$.

4. The unobserved effect a_i is eliminated.

Estimation and Interpretation

- The differenced equation is estimated by OLS using the cross-section of changes.
- $\hat{\beta}_1$ estimates the effect of a change in x on the change in y , holding a_i fixed.
- $\hat{\delta}_0$ (the intercept in the differenced regression) estimates the autonomous change in y over time.

5.3.4 Key Assumptions for FD Estimator Consistency

1. **Strict Exogeneity of u_{it} (conditional on a_i):** The original idiosyncratic errors u_{it} must be uncorrelated with x_{is} in *all* time periods ($s, t \in \{1, 2\}$). This implies $Cov(\Delta x_i, \Delta u_i) = 0$.
2. **Variation in Δx_i :** The change in x , Δx_i , must have non-zero variance across units. If x_{it} is time-constant, its effect cannot be estimated.
3. (For valid standard OLS inference): Homoskedasticity of Δu_i , i.e., $Var(\Delta u_i | \Delta x_i)$ is constant.

Assumption FD.1 (Model Structure)

We assume for each individual i , the model is:

$$y_{it} = \beta_1 x_{it1} + \cdots + \beta_k x_{itk} + a_i + u_{it}, \quad t = 1, \dots, T$$

where the β_j are the parameters to estimate and a_i is the unobserved effect.

Assumption FD.2 (Random Sampling)

We have a random sample from the cross section.

Assumption FD.3 (Time Variation and No Perfect Collinearity)

Each explanatory variable changes over time (for at least some i), and no perfect linear relationships exist among the explanatory variables.

For the next assumption, it is useful to let $\mathbf{X}_i$ denote the explanatory variables for all time periods for cross-sectional observation i ; thus, $\mathbf{X}_i$ contains $x_{itj}, t = 1, \dots, T, j = 1, \dots, k$.

Assumption FD.4 (Strict Exogeneity)

For each t , the expected value of the idiosyncratic error given the explanatory variables in *all* time periods and the unobserved effect is zero:

$$\mathbb{E}(u_{it} | \mathbf{X}_i, a_i) = 0.$$

When Assumption FD.4 holds, we sometimes say that the x_{itj} are *strictly exogenous conditional on the unobserved effect*. The idea is that, once we control for a_i , there is no correlation between the x_{isj} and the remaining idiosyncratic error, u_{it} , for all s and t .

As stated, Assumption FD.4 is stronger than necessary. We use this form of the assumption because it emphasizes that we are interested in the equation:

$$\mathbb{E}(y_{it} | \mathbf{X}_i, a_i) = \mathbb{E}(y_{it} | x_{it1}, \dots, x_{itk}, a_i) = \beta_1 x_{it1} + \cdots + \beta_k x_{itk} + a_i,$$

so that the β_j measure partial effects of the observed explanatory variables holding fixed, or “controlling for,” the unobserved effect a_i . Nevertheless, an important implication of FD.4, and one that is sufficient for the unbiasedness of the FD estimator, is:

$$\mathbb{E}(\Delta u_{it} | \mathbf{X}_i) = 0, \quad t = 2, \dots, T.$$

In fact, for consistency, we can simply assume that Δx_{itj} is uncorrelated with Δu_{it} for all $t = 2, \dots, T$ and $j = 1, \dots, k$.

Under these first four assumptions, the first-difference estimators are unbiased. The key assumption is FD.4, which is strict exogeneity of the explanatory variables. Under these same assumptions, we can also show that the FD estimator is consistent with a fixed T and as $N \rightarrow \infty$ (and perhaps more generally).

The next two assumptions ensure that the standard errors and test statistics resulting from pooled OLS on the first differences are (asymptotically) valid.

Assumption FD.5 (Homoskedasticity of Differenced Errors)

The variance of the differenced errors, conditional on all explanatory variables, is constant:

$$\text{Var}(\Delta u_{it} \mid \mathbf{X}_i) = \sigma^2, \quad t = 2, \dots, T.$$

Assumption FD.6 (No Serial Correlation of Differenced Errors)

For all $t \neq s$, the *differences* in the idiosyncratic errors are uncorrelated (conditional on all explanatory variables):

$$\text{Cov}(\Delta u_{it}, \Delta u_{is} \mid \mathbf{X}_i) = 0, \quad t \neq s.$$

Assumption FD.5 ensures that the differenced errors, Δu_{it} , are homoskedastic. Assumption FD.6 states that the differenced errors are serially uncorrelated, which means that the u_{it} follow a random walk across time (see Chapter 11). Under Assumptions FD.1 through FD.6, the FD estimator of the β_j is the best linear unbiased estimator (conditional on the explanatory variables).

Assumption FD.7(Normality)

Conditional on $\mathbf{X}_i$, the Δu_{it} are independent and identically distributed normal random variables.

When we add Assumption FD.7, the FD estimators are normally distributed, and the t - and F -statistics from pooled OLS on the differences have exact t and F distributions. Without FD.7, we can rely on the usual asymptotic approximations.

5.3.5 Advantages and Disadvantages of FD (for $T=2$)

- **Advantages:**
 - Controls for time-constant unobserved heterogeneity (a_i) potentially correlated with x_{it} .
 - Conceptually simple.
- **Disadvantages:**
 - Requires panel data.
 - Can reduce regressor variation, leading to imprecise estimates (large SEs).
 - Cannot estimate effects of time-invariant variables.
 - Can exacerbate measurement error bias.

5.3.6 Extension to Multiple Explanatory Variables

If the original model is $y_{it} = \beta_0 + \delta_0 d_{2t} + \beta_1 x_{it1} + \dots + \beta_k x_{itk} + a_i + u_{it}$, the differenced equation is:

$$\Delta y_i = \delta_0 + \beta_1 \Delta x_{i1} + \beta_2 \Delta x_{i2} + \dots + \beta_k \Delta x_{ik} + \Delta u_i$$

This is estimated by OLS.

6 Advanced Panel Data Methods

6.1 Fixed Effects Estimation

The goal is to eliminate the time-constant unobserved effect a_i .

6.1.1 Derivation

Consider the model for unit i at time t :

$$y_{it} = \beta_1 x_{it1} + \cdots + \beta_k x_{itk} + a_i + u_{it}, \quad t = 1, \dots, T \quad (3)$$

1. **Average over time for each unit i :** Let $\bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{it}$, $\bar{x}_{ij} = \frac{1}{T} \sum_{t=1}^T x_{itj}$, and $\bar{u}_i = \frac{1}{T} \sum_{t=1}^T u_{it}$. Since a_i is constant over time for unit i , $\frac{1}{T} \sum_{t=1}^T a_i = a_i$. The time-averaged equation is:

$$\bar{y}_i = \beta_1 \bar{x}_{i1} + \cdots + \beta_k \bar{x}_{ik} + a_i + \bar{u}_i \quad (4)$$

2. **Subtract the averaged equation from the original equation for each t :** $(y_{it} - \bar{y}_i) = \beta_1(x_{it1} - \bar{x}_{i1}) + \cdots + \beta_k(x_{itk} - \bar{x}_{ik}) + (a_i - a_i) + (u_{it} - \bar{u}_i)$
3. **The Time-Demeaned (or Within-Transformed) Equation:** Let $\ddot{y}_{it} = y_{it} - \bar{y}_i$, $\ddot{x}_{itj} = x_{itj} - \bar{x}_{ij}$, $\ddot{u}_{it} = u_{it} - \bar{u}_i$.

$$\ddot{y}_{it} = \beta_1 \ddot{x}_{it1} + \cdots + \beta_k \ddot{x}_{itk} + \ddot{u}_{it} \quad (5)$$

The unobserved effect a_i is eliminated.

Fixed Effects (FE) Estimator (or Within Estimator): Pooled OLS applied to the time-demeaned equation (5).

6.1.2 Key Characteristics and Assumptions of FE

- **Eliminates Time-Constant Variables:** Variables that do not vary over time for any unit are removed by demeaning ($\ddot{x}_{itj} = 0$ if x_{itj} is time-constant).
- **Allows $Cov(x_{itj}, a_i) \neq 0$:** This is a major strength. Because of this, any explanatory variable that is constant over time for all i gets swept away by the fixed effects transformation: $\ddot{x}_{it} = 0$ for all i and t , if x_{it} is constant across t . Therefore, we cannot include variables such as place of birth or a city's distance from a river.
- **Strict Exogeneity of u_{it} :** For unbiasedness/consistency, u_{it} must be uncorrelated with x_{isj} in *all* time periods s and t for a given unit i .
- For standard OLS inference on demeaned data to be valid, u_{it} are assumed homoskedastic and serially uncorrelated. Robust standard errors can be used otherwise.

6.1.3 Degrees of Freedom (df)

- For a balanced panel with N units and T time periods, and k explanatory variables in the demeaned model:

$$df = NT - N - k = N(T - 1) - k$$

One df is lost for each unit due to the demeaning process.

6.1.4 The Dummy Variable Regression (Alternative View of FE)

- Treats a_i as parameters to estimate by including a dummy variable for each unit i in the original model (plus time dummies and regressors x_{itj}).
- OLS on this dummy variable regression yields **identical** $\hat{\beta}_j$ and standard errors as the FE estimator from time-demeaning.
- Impractical for large N .
- Estimates of $\hat{a}_i$ can be recovered: $\hat{a}_i = \bar{y}_i - \sum_{j=1}^k \hat{\beta}_j \bar{x}_{ij}$. These are unbiased but not consistent as $N \rightarrow \infty$ with fixed T .

- The “intercept” often reported by FE software is $\frac{1}{N} \sum \hat{a}_i$. In other words, the overall intercept is actually the average of the individual-specific intercepts, which is an unbiased, consistent estimator of $\alpha = \mathbb{E}(a_i)$

6.1.5 Fixed Effects (FE) vs. First Differencing (FD)

- **If $T=2$:** FE and FD estimators are identical (with appropriate intercept handling).
- **If $T \geq 3$:**
 - **Efficiency:** Depends on serial correlation in u_{it} .
 - * If u_{it} are serially uncorrelated, FE is more efficient. Because the unobserved effects model is typically stated (sometimes only implicitly) with serially uncorrelated idiosyncratic errors, the FE estimator is used more than the FD estimator.
 - * If u_{it} follow a random walk, FD is more efficient.
 - **Strict Exogeneity Violations (e.g., lagged dependent variable):** FE bias tends to 0 as $T \rightarrow \infty$; FD bias does not. FE often preferred if $T > 2$.
- When T is large, and especially when N is not very large (for example, $N = 20$ and $T = 30$), we must exercise caution in using the fixed effects estimator. Although exact distributional results hold for any N and T under the classical fixed effects assumptions, inference can be very sensitive to violations of the assumptions when N is small and T is large.
 - First differencing has the advantage of turning an integrated time series process into a weakly dependent process. Therefore, if we apply first differencing, we can appeal to the central limit theorem even in cases where T is larger than N .

6.1.6 Fixed Effects with Unbalanced Panels

- **Unbalanced Panel:** Units observed for different numbers of time periods (T_i).
- **Mechanics:** Time-demeaning for unit i uses its own available T_i observations.
- Units with $T_i = 1$ (one observation) drop out (no within-unit variation).
- $df = (\sum_{i=1}^N T_i) - N - k$.
- **Critical Issue: Reason for Unbalance.**
 - If missingness is uncorrelated with u_{it} , FE is generally fine.
 - If attrition (units dropping out) is correlated with u_{it} , FE estimates can suffer from selection bias.
 - FE allows attrition to be correlated with the time-constant a_i .

For general missing data patterns, FE has an advantage over FD. Namely, FE maximizes the number of observations used in estimation, whereas FD only uses a time period t if time periods t and $t - 1$ both include a full set of observations.

6.2 Random Effects Models

The model is the same unobserved effects model:

$$y_{it} = \beta_0 + \beta_1 x_{it1} + \cdots + \beta_k x_{itk} + a_i + u_{it} \quad (6)$$

where a_i is the time-constant unobserved individual effect, and u_{it} is the idiosyncratic error.

6.2.1 The Crucial Random Effects Assumption

- The unobserved effect a_i is **uncorrelated** with all explanatory variables x_{itj} in all time periods:

$$\text{Cov}(x_{itj}, a_i) = 0 \quad \text{for all } t, j$$

- This is a stronger assumption than in Fixed Effects (FE).

6.2.2 The Composite Error Term ($v_{it} = a_i + u_{it}$)

If $Cov(x_{itj}, a_i) = 0$, a_i can be treated as part of the error. The model becomes: $y_{it} = \beta_0 + \sum_{j=1}^k \beta_j x_{itj} + v_{it}$.

Properties of v_{it} Assuming $E(a_i) = 0, E(u_{it}) = 0, Var(a_i) = \sigma_a^2, Var(u_{it}) = \sigma_u^2, Cov(a_i, u_{it}) = 0, Cov(u_{it}, u_{is}) = 0$ for $t \neq s$.

1. **Variance:**

$$Var(v_{it}) = Var(a_i + u_{it}) = Var(a_i) + Var(u_{it}) = \sigma_a^2 + \sigma_u^2$$

2. **Covariance (for $t \neq s$):**

$$Cov(v_{it}, v_{is}) = Cov(a_i + u_{it}, a_i + u_{is}) = Var(a_i) = \sigma_a^2$$

This implies v_{it} is serially correlated for a given individual i .

3. **Serial Correlation (for $t \neq s$):**

$$Corr(v_{it}, v_{is}) = \frac{Cov(v_{it}, v_{is})}{\sqrt{Var(v_{it})Var(v_{is})}} = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_u^2}$$

Implication for Pooled OLS: Consistent coefficients (if RE assumption holds) but incorrect standard errors and inefficient estimates due to ignoring serial correlation.

6.2.3 Estimation: Generalized Least Squares (GLS)

GLS is used to obtain efficient estimates.

The GLS Transformation (Quasi-Demeaning) Define the transformation parameter:

$$\theta = 1 - \sqrt{\frac{\sigma_u^2}{\sigma_u^2 + T\sigma_a^2}} \quad (7)$$

where T is the number of time periods. Note $0 \leq \theta \leq 1$. The transformed equation is:

$$(y_{it} - \theta \bar{y}_i) = \beta_0(1 - \theta) + \sum_{j=1}^k \beta_j(x_{itj} - \theta \bar{x}_{ij}) + (v_{it} - \theta \bar{v}_i) \quad (8)$$

OLS on this equation yields the GLS estimator. The transformed error $(v_{it} - \theta \bar{v}_i)$ is serially uncorrelated.

6.2.4 Feasible GLS (The Random Effects Estimator)

- Since σ_u^2 and σ_a^2 (and thus θ) are unknown, they are estimated from the data (e.g., using FE residuals for $\hat{\sigma}_u^2$).
- Using an estimated $\hat{\theta}$ in the transformation gives the **Random Effects (RE) estimator**.
- RE estimator is consistent and asymptotically normal as $N \rightarrow \infty$ (fixed T).

6.2.5 Advantages of Random Effects

- Can estimate coefficients on **time-constant explanatory variables** (unlike FE).
- More **efficient** than FE if the RE assumption $Cov(x_{itj}, a_i) = 0$ holds.

6.2.6 Relationship to Pooled OLS and Fixed Effects via θ

- If $\theta \approx 0$ (e.g., σ_a^2 small or T small), RE estimates $\approx$ Pooled OLS estimates.
- If $\theta \approx 1$ (e.g., σ_a^2 large or T large), RE estimates $\approx$ FE estimates.
- Bias from correlated a_i is attenuated by factor $(1 - \theta)$ in RE. As $\theta \rightarrow 1$ the bias term goes to zero, as it must because the RE estimator tends to the FE estimator. If θ is close to zero, we are leaving a larger fraction of the unobserved effect in the error term, and, as a consequence, the asymptotic bias of the RE estimator will be larger.

6.2.7 Random Effects vs. Pooled OLS

- **Breusch-Pagan Test:** Tests $H_0 : \sigma_a^2 = 0$. If rejected, RE or Pooled OLS with robust SEs preferred over simple Pooled OLS. Does not test $Cov(x_{itj}, a_i) = 0$.
- If RE assumptions hold, RE is more efficient than Pooled OLS.

6.2.8 Random Effects vs. Fixed Effects (The Key Decision)

- **Core Issue:** Is a_i correlated with any x_{itj} ?
 - If $Cov(x_{itj}, a_i) \neq 0 \implies$ RE is inconsistent; FE is consistent.
 - If $Cov(x_{itj}, a_i) = 0 \implies$ Both consistent; RE is more efficient.
- **Hausman Test:** Compares RE and FE coefficients on time-varying variables.
 - H_0 : Differences are not systematic ($Cov(x_{itj}, a_i) = 0$).
 - If H_0 is rejected, FE is preferred.
- Choice should be based on the plausibility of $Cov(x_{itj}, a_i) = 0$, not philosophical views of a_i .

6.2.9 Deeper Dive: Feasible Generalized Least Squares (FGLS) for Random Effects

The Random Effects (RE) model $y_{it} = \beta_0 + X_{it}\beta + a_i + u_{it}$ has a composite error $v_{it} = a_i + u_{it}$. For a given unit i , the vector of errors $\mathbf{v}_i = (v_{i1}, \dots, v_{iT})'$ has a covariance matrix:

$$\Omega_i = E(\mathbf{v}_i \mathbf{v}_i') = \sigma_u^2 I_T + \sigma_a^2 J_T$$

where I_T is a $T \times T$ identity matrix and J_T is a $T \times T$ matrix of ones. This non-spherical error structure ($\Omega_i \neq \sigma_v^2 I_T$) means Pooled OLS is inefficient and its standard errors are invalid.

Generalized Least Squares (GLS) - The Ideal: If Ω_i (and thus the full $NT \times NT$ block-diagonal error covariance matrix Ω) were known, the GLS estimator is BLUE:

$$\hat{\beta}_{GLS} = (\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{X}'\Omega^{-1}\mathbf{y}$$

This can be achieved by finding a transformation matrix P such that $P_i\Omega_iP_i' = \sigma^2 I_T$, and then applying OLS to the transformed model $P_i\mathbf{y}_i = P_i\mathbf{X}_i\beta + P_i\mathbf{v}_i$. The RE transformation $y_{it} - \theta\bar{y}_i$ is this operation.

Feasible GLS (FGLS) - The Practical Approach: Since Ω_i depends on unknown σ_a^2 and σ_u^2 , FGLS proceeds in two steps:

Step 1: Estimate σ_u^2 and σ_a^2 .

- **Estimating σ_u^2** (variance of idiosyncratic error): Use residuals from a Fixed Effects (FE) regression. Let $\hat{u}_{it}$ be the FE residuals.

$$\hat{\sigma}_u^2 = \frac{\sum_{i=1}^N \sum_{t=1}^T \hat{u}_{it}^2}{N(T-1) - k_{FE}}$$

(Denominator is df of FE: $NT - N - k_{FE}$ if an overall intercept is absorbed or $N(T-1) - k_{FE}$).

- **Estimating σ_a^2** (variance of individual effect): Commonly, from the residuals of a "Between Regression" ($\bar{y}_i$ on $\bar{X}_i$). The error in this regression is $\epsilon_i = a_i + \bar{u}_i$, with $Var(\epsilon_i) = \sigma_a^2 + \sigma_u^2/T$. Let $\hat{\sigma}_{BE}^2$ be the estimated variance of these residuals.

$$\hat{\sigma}_a^2 = \hat{\sigma}_{BE}^2 - \frac{\hat{\sigma}_u^2}{T}$$

Alternatively, using Pooled OLS residuals $\hat{v}_{it}$, $\hat{\sigma}_v^2 = \text{variance of } \hat{v}_{it}$. Then $\hat{\sigma}_a^2 = \hat{\sigma}_v^2 - \hat{\sigma}_u^2$. If $\hat{\sigma}_a^2 < 0$, it is typically set to 0.

Step 2: Perform GLS using estimated $\hat{\Omega}_i$. This involves calculating the estimated RE transformation parameter:

$$\hat{\theta} = 1 - \sqrt{\frac{\hat{\sigma}_u^2}{\hat{\sigma}_u^2 + T\hat{\sigma}_a^2}}$$

(If $\hat{\sigma}_a^2 = 0$, then $\hat{\theta} = 0$). Then, transform the data ("quasi-demeaning"):

$$\tilde{y}_{it} = y_{it} - \hat{\theta}\bar{y}_i$$

$$\tilde{x}_{itj} = x_{itj} - \hat{\theta}\bar{x}_{ij} \quad (\text{for each regressor } j)$$

The intercept becomes $\beta_0^* = \beta_0(1 - \hat{\theta})$

The FGLS estimator (the RE estimator) is obtained by OLS on the transformed model:

$$\tilde{y}_{it} = \beta_0^* + \beta_1\tilde{x}_{it1} + \cdots + \beta_k\tilde{x}_{itk} + \text{transformed error}$$

Mathematical Intuition for the θ Transformation: The transformation P_i that results in $y_{it} - \theta\bar{y}_i$ is constructed such that if it were applied to the true errors $\mathbf{v}_i$, the transformed errors $P_i\mathbf{v}_i$ would have a covariance matrix $\sigma_u^2 I_T$. This means the transformed errors are homoskedastic and serially uncorrelated. The specific formula for θ (equation 7) is derived from the structure of Ω_i^{-1} or by directly solving for a transformation that diagonalizes Ω_i . Essentially, the transformation optimally combines information from the within-unit variation (like FE) and the between-unit variation.

- If $T \rightarrow \infty$, $\theta \rightarrow 1$. RE approaches FE, as $\bar{y}_i$ becomes a perfect estimate of the unit-specific mean component involving a_i .
- If $\sigma_a^2 \rightarrow 0$, $\theta \rightarrow 0$. RE approaches Pooled OLS, as there's no individual random effect.

Properties of FGLS (RE Estimator):

- **Consistency:** Consistent if $\hat{\sigma}_u^2$ and $\hat{\sigma}_a^2$ are consistent estimators and standard exogeneity $E(X'_{it}v_{it}) = 0$ holds.
- **Asymptotic Normality:** Typically asymptotically normal.
- **Asymptotic Efficiency:** Achieves the same asymptotic variance as true GLS; more efficient than Pooled OLS if $\sigma_a^2 > 0$.
- **Finite Sample Properties:** Can be sensitive to the quality of $\hat{\sigma}_u^2, \hat{\sigma}_a^2$. Not always superior to OLS with robust SEs in small samples. Iteration (re-estimating Ω and β) does not change asymptotic properties if first-step estimates are consistent.